Linear Regression

Lecture Notes

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A.Y. 2026–2027

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1 Functional Predictors in Regression

In **regression** the target is a real number, $\mathcal{Y} = \mathbb{R}$. Since the prediction is already a value of the target, no decision rule is needed on top of a score: the predictor $h(\mathbf{x})$ is used directly, and the score-then-decision pipeline of classification collapses into a single stage. A loss $L(y, t)$ compares a prediction y with an observed target t, and the empirical risk is its average over the training set,

$$\overline{\mathcal{R}}_{\mathcal{T}}(h) = \frac{1}{|\mathcal{T}|} \sum_{(\mathbf{x},t)\in\mathcal{T}} L\big(h(\mathbf{x}), t\big). \tag{1}$$

As in the foundations notes, the relation between items and targets can be *functional*, with $t = f(\mathbf{x})$ for an unknown $f : \mathcal{X} \to \mathbb{R}$, or *probabilistic*, governed by an unknown conditional distribution $p_C(\cdot|\mathbf{x})$; with a functional predictor, the two hypotheses give the **FF** and **FP** cases. The functional hypothesis is strong: it requires that the features describe an item completely, and that they are measured without noise. The probabilistic one is more realistic, since relevant variables may be missing from $\mathbf{x}$, and the observed ones are often approximate. The functional case is the degenerate one in which $p_C(\cdot|\mathbf{x})$ puts all its mass on $f(\mathbf{x})$, so that the probabilistic case can be developed first.

In the FP case, the risk at $\mathbf{x}$ and the expected risk are

$$\mathcal{R}(\mathbf{x}; h) = \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})} \big[L(h(\mathbf{x}), t)\big], \qquad \mathcal{R}(h) = \mathop{\mathbb{E}}_{\mathbf{x}\sim p_M} \big[\mathcal{R}(\mathbf{x}; h)\big]. \tag{2}$$

Even a perfect model cannot predict t exactly. Adding and subtracting $L(h^B(\mathbf{x}), t)$ inside the expectation, where $h^B(\mathbf{x}) = \operatorname{argmin}_y \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})} [L(y, t)]$ is the Bayes predictor, the risk at $\mathbf{x}$ splits as

$$\mathcal{R}(\mathbf{x}; h) = \underbrace{D\big(p_C(\cdot|\mathbf{x}), h(\mathbf{x})\big)}_{\text{reducible}} + \underbrace{C\big(p_C(\cdot|\mathbf{x})\big)}_{\text{irreducible}}, \tag{3}$$

with

$$D\big(p_C(\cdot|\mathbf{x}), y\big) = \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})} \big[L(y, t) - L(h^B(\mathbf{x}), t)\big],$$
$$C\big(p_C(\cdot|\mathbf{x})\big) = \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})} \big[L(h^B(\mathbf{x}), t)\big].$$

The **reducible** risk is non-negative and vanishes at $y = h^B(\mathbf{x})$: it measures what a better predictor could still remove. The **irreducible** risk, or *noise*, depends only on $p_C(\cdot|\mathbf{x})$ and on the loss, never

on h. Since the target is continuous, the expectations are integrals, and explicit expressions are available only for specific losses, such as those of Sections 4 and 15.

The reducible risk depends on the random training set through the learned predictor $h_{\mathcal{T}}$. Averaging over training sets, with *main prediction* $\bar{h}(\mathbf{x}) = \operatorname{argmin}_z \mathbb{E}_{\mathcal{T}}[L(h_{\mathcal{T}}(\mathbf{x}), z)]$, gives the bias–variance decomposition of the foundations notes,

$$\mathop{\mathbb{E}}_{\mathcal{T}}[\mathcal{R}(\mathbf{x}; h_{\mathcal{T}})] = \underbrace{C\big(p_C(\cdot|\mathbf{x})\big)}_{\text{irreducible}} + \underbrace{D\big(p_C(\cdot|\mathbf{x}), \bar{h}(\mathbf{x})\big)}_{\text{bias}} + \underbrace{\mathop{\mathbb{E}}_{\mathcal{T}}\big[\mathcal{R}(\mathbf{x}; h_{\mathcal{T}}) - \mathcal{R}(\mathbf{x}; \bar{h})\big]}_{\text{variance}},$$

an identity valid for every loss. Bias and variance have their intuitive meaning, the bias depending only on where $\bar{h}(\mathbf{x})$ lies and the variance only on the spread of $h_{\mathcal{T}}(\mathbf{x})$, only for some losses, among which the quadratic loss.

The empirical risk (1) has the same form in the FP and FF cases, but in FP each t_i is now a random draw from $p_C(\cdot|\mathbf{x}_i)$ rather than exactly $f(\mathbf{x}_i)$. Since the pairs are drawn independently from $p_M(\mathbf{x})\, p_C(t|\mathbf{x})$, by the law of large numbers $\overline{\mathcal{R}}_{\mathcal{T}}(h) \to \mathcal{R}(h)$ for every fixed h, and the considerations given on learning as the minimization of $\overline{\mathcal{R}}_{\mathcal{T}}(h)$ over $h \in \mathcal{H}$ apply.

Empirical distribution and Dirac delta

A finite collection of observed values $t_1, \dots, t_n$, possibly with repetitions, defines a probability distribution $\hat{p}$, the **empirical distribution**, which gives non-zero probability only to the observed values, each in proportion to the number of times it appears: it describes the data as if they were the whole population. An expectation under $\hat{p}$ is simply the average over the observed values,

$$\mathop{\mathbb{E}}_{u\sim\hat{p}}[g(u)] = \frac{1}{n}\sum_{i=1}^{n} g(t_i),$$

which, by the law of large numbers, approaches the expectation under the distribution that generated the values as n grows. The same construction applies to a training set of pairs $(\mathbf{x}_i, t_i)$, whose empirical distribution $\hat{p}_{\mathcal{T}}$ gives probability $1/|\mathcal{T}|$ to each pair: replacing p_J with $\hat{p}_{\mathcal{T}}$ turns the risk into the empirical risk.

Classification (finite $\mathcal{Y}$). $\hat{p}$ is a probability mass function, which gives each class its relative frequency, $\hat{p}(k) = \frac{1}{n}\big|\{i : t_i = k\}\big|$.
Regression ($\mathcal{Y} = \mathbb{R}$). $\hat{p}$ concentrates a mass $\frac{1}{n}$ on each observed value, and cannot be written as an ordinary density. It is written with the *Dirac delta* δ, a generalized density which is zero everywhere except at 0 and has integral 1. It is not a function in the usual sense, but it is defined by its effect on integrals, $\int g(u)\,\delta(u-t)\,du = g(t)$: the integral of a function $g(u)$ multiplied by a Dirac delta centered at t returns the value $g(t)$. Then

$$\hat{p}(u) = \frac{1}{n}\sum_{i=1}^{n} \delta(u - t_i),$$

and integrating g against $\hat{p}$ gives the average above.

Notation. We write δ_t for the empirical distribution of a single value t, which concentrates all the probability on t. In classification it is the probability vector with $\delta_t(k) = 1$ for $k = t$ and 0 otherwise (the Kronecker delta, or the one-hot vector of class t); in regression it is the Dirac delta centered at t, $\delta_t(u) = \delta(u - t)$. In both cases $\mathbb{E}_{u\sim\delta_t}[g(u)] = g(t)$.

The FF case is recovered considering $p_C(t|\mathbf{x}) = \delta_{f(\mathbf{x})}(t)$, the empirical distribution concentrated at $f(\mathbf{x})$: expectations over t become evaluations at $t = f(\mathbf{x})$. The risk at $\mathbf{x}$ is $L(h(\mathbf{x}), f(\mathbf{x}))$; for any loss with $L(z, z) = 0$ and $L(y, z) > 0$ for $y \neq z$, such as the quadratic, absolute and Huber losses, the Bayes predictor is f itself and the irreducible risk vanishes, so that the whole risk is

reducible, and bias and variance account for the whole expected error. The minimum risk over all functions is zero, attained by f, but whether learning can reach it depends on **realizability**, that is, on whether the hypothesis class $\mathcal{H}$ contains a function agreeing with f on all of $\mathcal{X}$, a property of the pair $(\mathcal{H}, f)$ independent of the loss.

2 Linear Models

Section 1 cast learning a regression model as the minimization of the empirical risk (1) over a hypothesis class $\mathcal{H}$, whose best members approximate the Bayes predictor h^B: the conditional mean of the target under the quadratic loss, as shown in Section 4. The choice of $\mathcal{H}$ determines both the approximation that can be attained and the difficulty of the minimization. The simplest and most widely used classes are the **linear models**, which predict the target as a linear combination of the input features. For a d-dimensional input $\mathbf{x} = (x_1, \ldots, x_d)^T$,

$$h(\mathbf{x}; \mathbf{w}, b) \;=\; w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + b \;=\; \mathbf{w}^T \mathbf{x} + b, \tag{4}$$

with **weights** $\mathbf{w} \in \mathbb{R}^d$ and a constant term b, the **bias**, which allows a non-zero prediction when all features are zero, shifting the hyperplane $h(\mathbf{x}) = 0$ away from the origin.

These models are linear in two distinct senses. They are linear in the *features* $\mathbf{x}$: the predictions lie on a hyperplane in input space. More importantly for learning, they are linear in the *parameters* $\mathbf{w}$ and b, the variables of the optimization: as shown in Section 5, this makes the sum of squared errors a convex function of the parameters, with a single global minimum that can be computed in closed form. The first kind of linearity can be removed without losing the second, by transforming the inputs before combining them, as discussed next.

3 Basis Functions

3.1 Representation

How the items are represented shapes everything that follows: a poor representation forces the model to work harder, and often makes it fail; a good one makes the structure of the data visible and learning tractable. Representations are often chosen on the basis of domain knowledge, by examining what the features measure and designing transformations accordingly. In neural networks, discussed later in the course, the model instead learns its own representation from the data. The polynomial regression example of Section 6 makes the role of representation concrete: the degree of the polynomial is a choice of representation, and its consequences for underfitting and overfitting are immediate.

A linear model is restricted to linear relationships in the input space. Its expressive power is extended by **feature engineering**: the input $\mathbf{x} \in \mathbb{R}^d$ is mapped into a new space $\mathbb{R}^m$ through a set of fixed **basis functions** $\phi_1, \ldots, \phi_m$, and the model becomes

$$h(\mathbf{x}; \mathbf{w}, b) \;=\; \sum_{j=1}^{m} w_j \phi_j(\mathbf{x}) + b \;=\; \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}) + b, \qquad \boldsymbol{\phi}(\mathbf{x}) = \big(\phi_1(\mathbf{x}), \ldots, \phi_m(\mathbf{x})\big)^T. \tag{5}$$

A relation that is non-linear in $\mathbb{R}^d$ can be linear in $\mathbb{R}^m$. The model is still a linear combination, with respect to $\mathbf{w}$, of fixed functions: only what is combined has changed, and linearity in the parameters, with all its consequences, is preserved.

Basis functions fall into two broad categories. **Domain-specific** basis functions encode knowledge about what the features measure: periodic functions for a feature representing time, radially symmetric ones for a feature encoding a distance. **General-purpose** basis functions make no assumption on the meaning of the features, and apply to any dataset because they are flexible enough to approximate a wide range of functions.

3.2 Common Basis Functions

Four general-purpose families are in wide use, shown in Figure 1 for a scalar input. They differ in how a change in a single weight w_j affects the predictions: everywhere in the input space, or only in a region.

- **Polynomial** basis functions, $\phi_j(x) = x^j$, are *global*: each of them varies over the whole input range, so a change in a single weight affects all predictions. They are flexible, but tend to behave erratically near the boundaries of the data.
- **Gaussian** basis functions, bell-shaped around a center μ_j with width s,

$$\phi_j(x) = \exp\left(-\frac{(x-\mu_j)^2}{2s^2}\right),$$

  are *local*: they are significantly different from zero only in a neighborhood of μ_j, so that each weight affects the predictions only there.
- **Sigmoid** basis functions produce a smooth step from 0 to 1 around μ_j,

$$\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right) = \frac{1}{1+e^{-(x-\mu_j)/s}}.$$

- **Hyperbolic tangent** basis functions produce the same step, from -1 to 1:

$$\phi_j(x) = 2\sigma\left(\frac{x-\mu_j}{s}\right) - 1 = \tanh\left(\frac{x-\mu_j}{2s}\right).$$

Sigmoid and hyperbolic tangent functions lie between the two extremes: they vary only in a neighborhood of μ_j, but, unlike Gaussians, they do not vanish away from it, being constant on each side of the step.

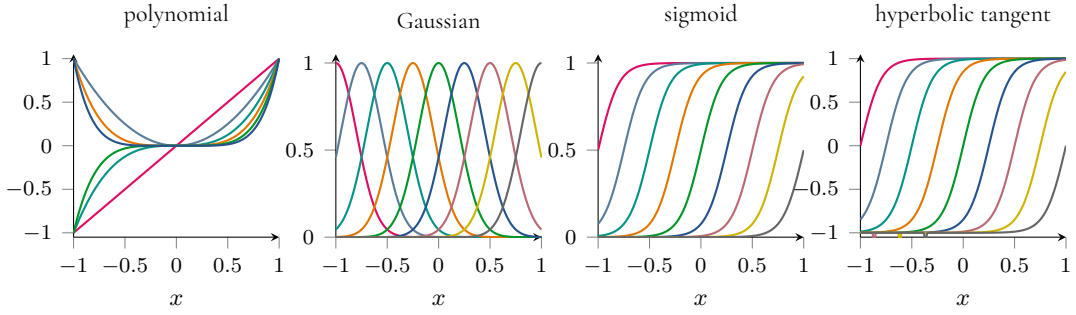


Figure 1: The four families of basis functions on $[-1, 1]$: polynomial x^j for $j = 1, \ldots, 6$ (global), Gaussian with $s = 0.2$ (local), sigmoid with $s = 0.1$ and hyperbolic tangent (steps whose variation is concentrated around their center μ_j), with centers μ_j evenly spaced in $[-1, 1]$.

3.3 The Design Matrix

A training set of n items is arranged as the matrix $\mathbf{X}$, whose i-th row is $\mathbf{x}_i^T$, and the vector of targets $\mathbf{t} = (t_1, \ldots, t_n)^T$. Applying $\boldsymbol{\phi}$ to every item gives the **design matrix** $\boldsymbol{\Phi}$, whose i-th row

is the transformed item $\boldsymbol{\phi}(\mathbf{x}_i)^T$ and whose j-th column is the basis function ϕ_j evaluated on all items:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}, \qquad \boldsymbol{\Phi} = \begin{pmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \cdots & \phi_m(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_n) & \phi_2(\mathbf{x}_n) & \cdots & \phi_m(\mathbf{x}_n) \end{pmatrix}. \tag{6}$$

In the following we refer to the training set $(\mathbf{X}, \mathbf{t})$ for brevity: all derivations apply unchanged to $(\boldsymbol{\Phi}, \mathbf{t})$, since the optimization does not depend on whether the features are the original ones or the result of a transformation.

Example: noisy observations of a sine wave

Consider n observations $(x_1, t_1), \ldots, (x_n, t_n)$ of a scalar input and a scalar target (Figure 2), generated by the relation

$$t_i = \sin(2\pi x_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

which is unknown to the learner. The aim is not to fit the noise, but to recover $\sin(2\pi x)$, so as to predict t for new inputs, from the noisy observations only. **Polynomial regression** models the relation with a polynomial of degree M,

$$h(x; \mathbf{w}, b) = \sum_{j=1}^{M} w_j x^j + b,$$

a linear model in the basis $\phi_j(x) = x^j$. The degree governs the balance between underfitting and overfitting: with $M = 1$ the model is a straight line, unable to follow the oscillations of the sine wave; with M large, it can follow every noisy observation. Choosing M is a first instance of **model selection**, discussed in Section 6.

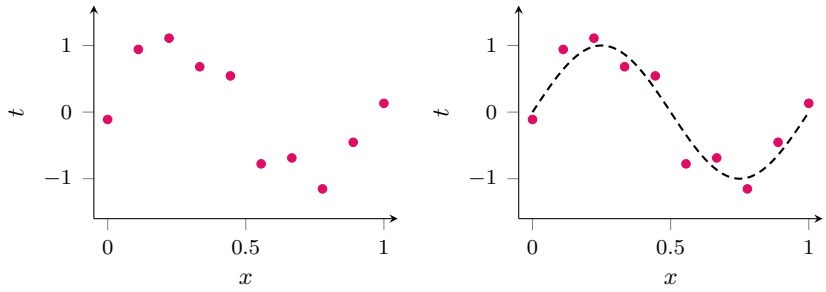


Figure 2: Left: a small training set of ten observations, a smooth trend buried under noise. Right: the same observations, sampled from the sine wave $\sin(2\pi x)$ (dashed) with Gaussian noise of standard deviation 0.3.

4 The Quadratic Loss

The parameters of a linear model are chosen by minimizing the empirical risk (1), and the standard choice of the loss in regression is the **quadratic loss**, or squared error,

$$L(y, t) = (y - t)^2. \tag{7}$$

Doubling an error quadruples its cost: small errors contribute very little, while a few large errors can dominate the total. The loss is convex and differentiable in y, and its gradient $2(y-t)$ grows linearly

with the residual (Figure 3): a badly fitted item produces a large corrective step, and can move the parameters more than many well-fitted ones, which explains the sensitivity of the quadratic loss to outliers. Over the training set it gives the **mean squared error**

$$\overline{\mathcal{R}}_{\mathcal{T}}(h) \;=\; \frac{1}{|\mathcal{T}|} \sum_{(\mathbf{x},t)\in\mathcal{T}} \big(h(\mathbf{x}) - t\big)^2, \tag{8}$$

which, for a model linear in its parameters, is a convex quadratic function of them, minimized in closed form in Section 5.

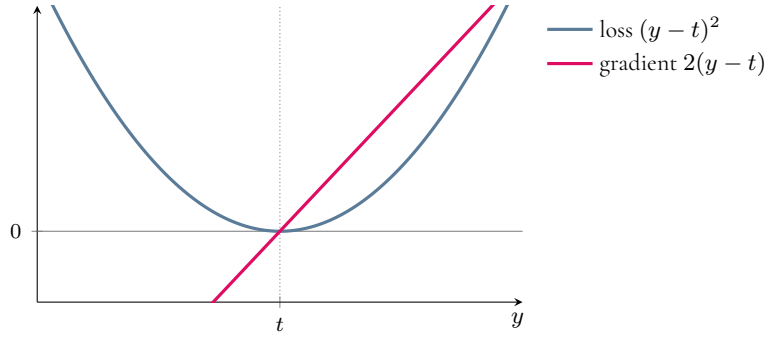


Figure 3: Quadratic loss and its gradient, which grows linearly with the error.

4.1 The Optimal Predictor

For the quadratic loss, the split (3) has an explicit form. Let $\mu(\mathbf{x}) = \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})}[t]$ and $\sigma^2(\mathbf{x}) = \mathop{Var}_{t\sim p_C(\cdot|\mathbf{x})}[t]$ be the conditional mean and variance of the target. Adding and subtracting $\mu(\mathbf{x})$ inside the square, the cross term vanishes, since $\mu(\mathbf{x})$ is the mean of t, and the risk at $\mathbf{x}$ is

$$\mathcal{R}(\mathbf{x};h) \;=\; \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})}\big[(h(\mathbf{x})-t)^2\big] \;=\; \underbrace{\big(h(\mathbf{x})-\mu(\mathbf{x})\big)^2}_{\text{reducible}} + \underbrace{\sigma^2(\mathbf{x})}_{\text{irreducible}}. \tag{9}$$

The reducible risk $D = (h(\mathbf{x})-\mu(\mathbf{x}))^2$ is minimized by the Bayes predictor $h^B(\mathbf{x}) = \mu(\mathbf{x})$, predicting the **conditional mean** and the irreducible one $C = \sigma^2(\mathbf{x})$ is the noise at $\mathbf{x}$, which vanishes in the FF case: learning a regression model amounts to approximating $\mu(\mathbf{x})$ within a hypothesis class. For a predictor $h_{\mathcal{T}}$ learned from a random training set, with average prediction $\bar{h}(\mathbf{x}) = \mathop{\mathbb{E}}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})]$[1], the bias–variance decomposition takes the familiar form

$$\mathop{\mathbb{E}}_{\mathcal{T}}[\mathcal{R}(\mathbf{x};h_{\mathcal{T}})] \;=\; \underbrace{\sigma^2(\mathbf{x})}_{\text{noise}} + \underbrace{\big(\bar{h}(\mathbf{x})-\mu(\mathbf{x})\big)^2}_{\text{bias}} + \underbrace{\mathop{Var}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})]}_{\text{variance}}, \tag{10}$$

[1] This is the main prediction $\bar{h}(\mathbf{x}) = \operatorname{argmin}_z \mathop{\mathbb{E}}_{\mathcal{T}}[L(h_{\mathcal{T}}(\mathbf{x}),z)]$ of Section 1, specialized to the quadratic loss. Writing $m(\mathbf{x}) = \mathop{\mathbb{E}}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})]$, adding and subtracting $m(\mathbf{x})$ inside the square gives

$$\begin{aligned}
\mathop{\mathbb{E}}_{\mathcal{T}}\Big[\big(h_{\mathcal{T}}(\mathbf{x})-z\big)^2\Big] &= \mathop{\mathbb{E}}_{\mathcal{T}}\Big[\big((h_{\mathcal{T}}(\mathbf{x})-m(\mathbf{x}))+(m(\mathbf{x})-z)\big)^2\Big] \\
&= \mathop{\mathbb{E}}_{\mathcal{T}}\Big[\big(h_{\mathcal{T}}(\mathbf{x})-m(\mathbf{x})\big)^2\Big] + 2\big(m(\mathbf{x})-z\big)\mathop{\mathbb{E}}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})-m(\mathbf{x})] + \big(m(\mathbf{x})-z\big)^2 \\
&= \mathop{Var}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})] + \big(m(\mathbf{x})-z\big)^2,
\end{aligned}$$

since $m(\mathbf{x})-z$ does not depend on $\mathcal{T}$ and can be taken out of the expectation, and $\mathop{\mathbb{E}}_{\mathcal{T}}[h_{\mathcal{T}}-m(\mathbf{x})] = m(\mathbf{x})-m(\mathbf{x}) = 0$, so that the cross term vanishes. The first term does not depend on z, and the second is non-negative and vanishes only for $z = m(\mathbf{x})$: the minimizer is $\bar{h}(\mathbf{x}) = m(\mathbf{x}) = \mathop{\mathbb{E}}_{\mathcal{T}}[h_{\mathcal{T}}(\mathbf{x})]$. The mean of the predictions is the value closest to them in expected squared distance, as the conditional mean is for the target. With other losses the main prediction is a different statistic of the predictions, such as their median for the absolute loss.

which will guide the discussion of regularization in Section 7.

The conditional mean weights every value of t by its probability. For a symmetric, unimodal and light-tailed $p_C(\cdot|\mathbf{x})$, such as a Gaussian, mean, median and mode coincide, and the mean is a good summary of the typical outcome. For a skewed or heavy-tailed distribution, instead, a distant tail with little mass can pull the mean away from the bulk of the distribution (Figure 4): the fit is then driven by rare extreme values, and the losses of Section 15 may be preferable. The assumption on the noise implicit in the quadratic loss is made explicit in Section 10.

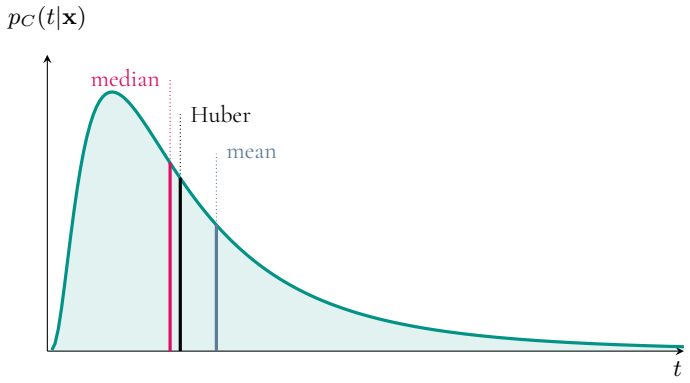


Figure 4: A right-skewed conditional distribution at a fixed $\mathbf{x}$, with the optimal predictors of the quadratic loss (mean), of the absolute loss (median) and of the Huber loss (between the two), discussed in Section 15. For a symmetric distribution the three coincide.

5 Least Squares and the Normal Equations

5.1 Derivation

Under the quadratic loss, the empirical risk of a linear model is proportional to the sum of squared residuals,

$$E(\mathbf{w}, b) \;=\; \sum_{i=1}^{n} r_i^2, \qquad r_i \;=\; \sum_{j=1}^{d} w_j x_{ij} + b - t_i, \tag{11}$$

where r_i is the residual of the i-th item, the difference between prediction and target. Each squared residual is a convex function of the parameters, and so is their sum: E has no local minima other than the global one, which is found by setting the gradient to zero. Since E is quadratic, its gradient is linear in the parameters,

$$\frac{\partial E}{\partial w_k} = 2\sum_{i=1}^{n} r_i \frac{\partial r_i}{\partial w_k} \;=\; 2\sum_{i=1}^{n}\Big(\sum_{j=1}^{d} w_j x_{ij} + b - t_i\Big) x_{ik}, \qquad k = 1, \dots, d,$$

$$\frac{\partial E}{\partial b} = 2\sum_{i=1}^{n} r_i \frac{\partial r_i}{\partial b} \;=\; 2\sum_{i=1}^{n}\Big(\sum_{j=1}^{d} w_j x_{ij} + b - t_i\Big),$$

and setting it to zero gives a system of $d+1$ linear equations.

The system is written compactly by absorbing the bias into the weights. The augmented weight vector $\overline{\mathbf{w}} = (b, w_1, \dots, w_d)^T \in \mathbb{R}^{d+1}$ and the augmented design matrix $\overline{\mathbf{X}}$, obtained by prepending a column of ones to $\mathbf{X}$,

$$\overline{\mathbf{X}} \;=\; \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{pmatrix}, \tag{12}$$

treat the bias as the weight of an input always equal to 1. The vector of predictions is then $\overline{\mathbf{X}}\overline{\mathbf{w}}$, the error is $E(\overline{\mathbf{w}}) = ||\overline{\mathbf{X}}\overline{\mathbf{w}} - \mathbf{t}||^2$, and setting its gradient to zero gives

$$\overline{\mathbf{X}}^T(\overline{\mathbf{X}}\overline{\mathbf{w}} - \mathbf{t}) = \mathbf{0}, \qquad \text{that is,} \qquad \overline{\mathbf{X}}^T\overline{\mathbf{X}}\,\overline{\mathbf{w}} = \overline{\mathbf{X}}^T\mathbf{t}. \tag{13}$$

When $\overline{\mathbf{X}}^T\overline{\mathbf{X}}$ is non-singular, which holds unless some features are linear combinations of the others, the unique solution is given by the **normal equations**

$$\overline{\mathbf{w}}^* = (\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}\overline{\mathbf{X}}^T\mathbf{t}. \tag{14}$$

The globally optimal parameters are obtained in a single step, with no iteration, step size or convergence check, at the cost of solving a $(d+1) \times (d+1)$ linear system, $O(d^3)$. This makes the normal equations the method of choice for moderate d; for large d, the iterative methods discussed in the notes on optimization are preferable.

Example: the normal equations at work

Take $n = 3$ items with a single feature, $x = 1, 2, 3$, and targets $t = 2, 3, 5$. Then

$$\overline{\mathbf{X}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \overline{\mathbf{X}}^T\overline{\mathbf{X}} = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad \overline{\mathbf{X}}^T\mathbf{t} = \begin{pmatrix} 10 \\ 23 \end{pmatrix},$$

so that $(\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$ and

$$\overline{\mathbf{w}}^* = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}\begin{pmatrix} 10 \\ 23 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 2 \\ 9 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 3/2 \end{pmatrix}:$$

the fitted model is $h(x) = 1.5\,x + 0.33$.

6 Overfitting and Model Selection

In the example of Section 3, the polynomial of degree M uses the basis functions $1, x, x^2, \ldots, x^M$, where $\phi_0(x) = 1$ makes the bias the weight w_0. The degree controls the number of coefficients to be estimated, and with it the flexibility of the curve (Figure 5). For $M = 0$ or $M = 1$ the model is too rigid to follow the oscillations of the sine wave. For $M+1 = n$ it has as many parameters as data points, and passes through every training point, with zero training error: it **overfits**, reproducing the noise of the training set instead of the underlying signal, and generalizes poorly.

The effect is measured by evaluating each model on a test set, generated by the same process as the training set but independently of it, as discussed in the foundations notes. The error is usually reported as the **root mean square error**,

$$E_{\text{RMS}} = \sqrt{\mathcal{R}_{\mathcal{T}'}(\overline{\mathbf{w}}^*)} = \sqrt{\frac{1}{|\mathcal{T}'|}\sum_{(x,t)\in\mathcal{T}'}\Big(\sum_{j=0}^{M} w_j^* x^j - t\Big)^2}, \tag{15}$$

on the scale of the target, where $\mathcal{T}'$ is the test set and $\overline{\mathbf{w}}^*$ the parameters learned on the training set. Figure 6 shows the two typical curves. The training error decreases monotonically with M, since a more flexible model always fits the training points at least as well. The test error first decreases, as the model captures the real structure of the data, and then increases, as it starts capturing the noise: the best degree is at its minimum.

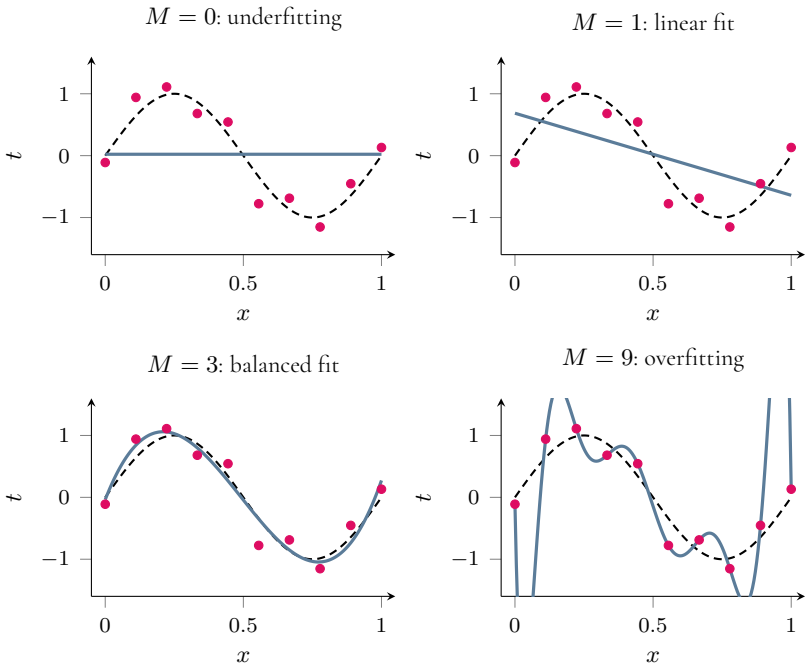


Figure 5: Polynomial fits (blue) to ten noisy observations (red) of $\sin(2\pi x)$ (dashed), for four values of the degree M. The curve with $M = 9$ passes through every point, but is useless for prediction.

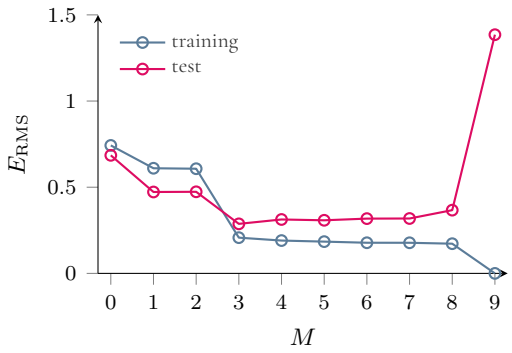


Figure 6: Training error (blue) and test error (red), computed on 100 independent test points, as functions of the degree M. The training error decreases monotonically, down to zero for $M = 9$, while the test error reaches its minimum at a moderate degree and then rises sharply.

7 Regularization

7.1 Controlling Complexity

Overfitting is not a matter of model complexity alone, but of the balance between the degrees of freedom of the model and the information in the data. For a fixed model, the tendency to overfit decreases as the training set grows, since a flexible curve must satisfy more constraints and has less room to follow the noise (Figure 7): larger training sets tolerate more complex models.

Data, however, are often scarce or expensive, and three approaches address the imbalance: collecting more data, choosing a smaller hypothesis class, such as a lower degree or fewer basis functions, and **regularization**, which keeps the class fixed but discourages extreme solutions. Regularization can impose a *hard constraint*, restricting the parameters to a bounded region, which complicates the optimization, or a *soft constraint*, adding to the objective a penalty on large parameter values.

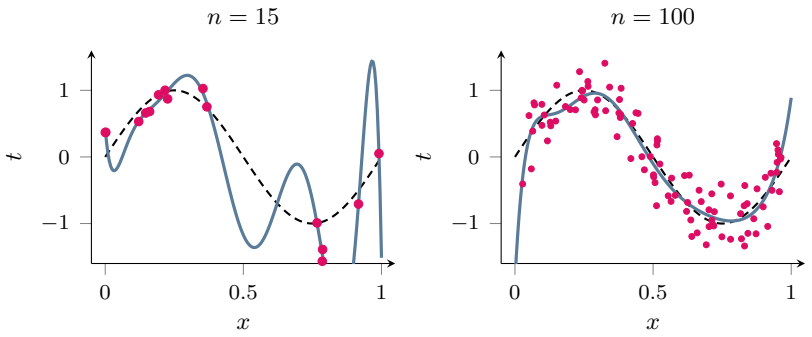


Figure 7: The same polynomial with $M = 9$ fitted on a small ($n = 15$) and on a large ($n = 100$) training set. More data force the curve to follow the signal rather than the noise.

The second preserves the smooth structure of the objective, and is the one developed here:

$$E(\overline{\mathbf{w}}) \;=\; E_D(\overline{\mathbf{w}}) + \lambda E_W(\overline{\mathbf{w}}). \tag{16}$$

The **data term** E_D is the empirical risk, up to a factor, and measures the fit to the training data; the **regularization term** E_W penalizes extreme parameter values; the **regularization coefficient** λ balances the two. A small λ favors the fit to the data, with the risk of overfitting; a large λ favors small weights, with the risk of underfitting.

7.2 Ridge Regression

The most common regularization term penalizes the squared norm of the weights. In the augmented notation, with $w_0 = b$,

$$E_W(\overline{\mathbf{w}}) \;=\; \frac{1}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}} \;=\; \frac{1}{2}\sum_{j=0}^{d} w_j^2, \tag{17}$$

where the bias is penalized together with the other weights, a convention that keeps the formulas compact; in practice the bias is often left out of the penalty, so that the predictions are not pulled toward zero. The objective is

$$E(\overline{\mathbf{w}}) \;=\; \frac{1}{2}\sum_{i=1}^{n} r_i^2 + \frac{\lambda}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}} \;=\; \frac{1}{2}\boldsymbol{r}^T\boldsymbol{r} + \frac{\lambda}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}}, \qquad \boldsymbol{r} = \overline{\mathbf{X}}\overline{\mathbf{w}} - \mathbf{t}, \tag{18}$$

and setting its gradient to zero gives the **regularized normal equations**

$$\overline{\mathbf{w}}^* \;=\; (\lambda\mathbf{I} + \overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}\overline{\mathbf{X}}^T\mathbf{t}. \tag{19}$$

The term $\lambda\mathbf{I}$ strengthens the diagonal before inversion, so that the matrix is well conditioned even when the features are highly correlated. This approach is called **ridge regression**, or Tikhonov regularization.

7.3 Lasso and Sparsity

More generally, the exponent of the penalty can be treated as a parameter q:

$$E(\overline{\mathbf{w}}) \;=\; \frac{1}{2}\sum_{i=1}^{n} r_i^2 + \frac{\lambda}{2}\sum_{j=0}^{d} |w_j|^q. \tag{20}$$

The case $q = 1$ is the **Lasso** (least absolute shrinkage and selection operator). Unlike ridge regression, which shrinks the weights toward zero but rarely makes them exactly zero, the Lasso often

returns sparse solutions, with many weights equal to zero: it performs a form of feature selection, ignoring the irrelevant features.

The reason is geometric. Minimizing Equation (16) is equivalent to minimizing $E_D(\overline{\mathbf{w}})$ under a constraint $E_W(\overline{\mathbf{w}}) \leq \eta$, with η decreasing as λ grows. In two dimensions, the ridge constraint $w_1^2 + w_2^2 \leq \eta$ is a disk, while the Lasso constraint $|w_1| + |w_2| \leq \eta$ is a square rotated by 45 degrees, with its vertices on the coordinate axes. The level curves of E_D are ellipses centered at the unconstrained least-squares solution, and the constrained solution is the point where the smallest ellipse touches the constraint region. For the disk, the contact point is generically one where both coordinates are non-zero; for the square, the ellipses tend to touch it first at a vertex, where one coordinate is zero (see Figure 8). The same happens in higher dimensions, where the vertices and edges of the constraint region correspond to solutions with several zero weights. Choosing between $q = 2$ and $q = 1$ is thus a design decision: a model that uses all inputs with small weights, or one that selects the most relevant ones.

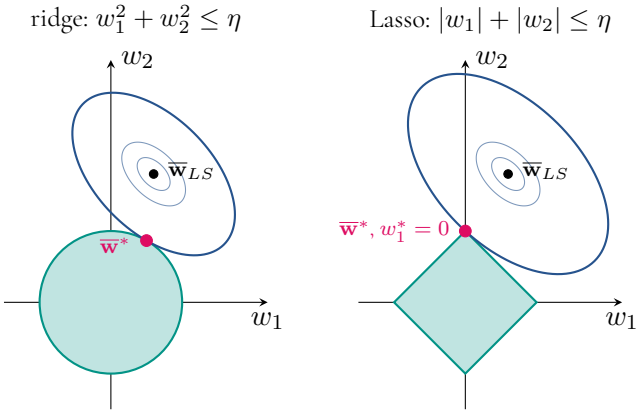


Figure 8: Ridge (left) and Lasso (right) as constrained minimization. The level curves of E_D are ellipses centered at the least-squares solution $\overline{\mathbf{w}}_{LS}$; the constrained solution $\overline{\mathbf{w}}^*$ is the point where the smallest ellipse touches the constraint region. The disk is touched at a point with both coordinates non-zero, the square at a vertex, where $w_1^* = 0$.

Example: the effect of λ

For the polynomial of degree $M = 9$, ridge regularization limits the magnitude of the coefficients, preventing the sharp oscillations needed to follow the noise (Figure 9). Three regimes emerge. For a small λ the penalty is negligible, the training error is small and the test error large (overfitting). For a large λ the coefficients are suppressed and the model is nearly flat, with large errors on both sets (underfitting). For an intermediate λ the test error is minimal (Figure 10).

The same effect can be observed across training sets. Consider $L = 100$ independent training sets of $n = 25$ points drawn from $\sin(2\pi x)$, and fit to each a model with 24 Gaussian basis functions, deliberately many, with ridge regularization (Figure 11).

- With strong regularization ($\ln \lambda = 2.6$), the 100 fitted curves are nearly identical, *low variance*, but their average is a poor approximation of the sine wave, *high bias*: the model is too rigid to learn the signal.
- With intermediate regularization ($\ln \lambda = -0.31$), the curves follow the sinusoidal trend, and their average fits it closely.
- With weak regularization ($\ln \lambda = -2.4$), each curve follows the noise of its own training set, and the curves are very different from each other, *high variance*, although their average remains close to the sine wave, *low bias*: any single model is unreliable.

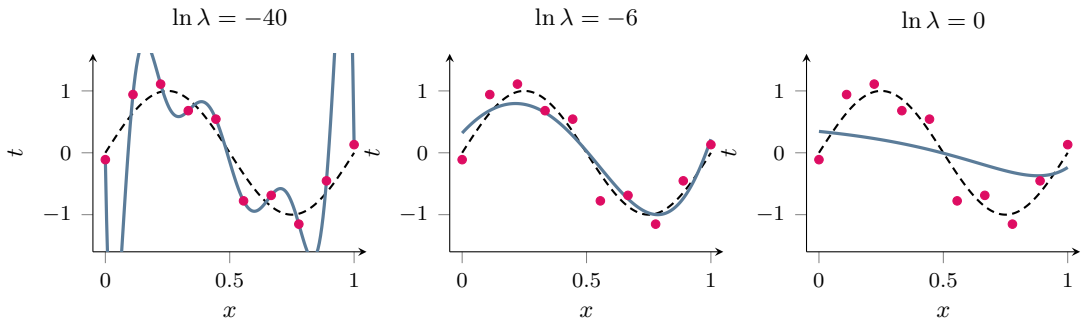


Figure 9: Effect of λ on a polynomial fit with $M = 9$ on the ten observations of Figure 5: negligible penalty ($\ln \lambda = -40$, overfitting), intermediate penalty ($\ln \lambda = -6$, close to the minimum of the test error), strong penalty ($\ln \lambda = 0$, underfitting).

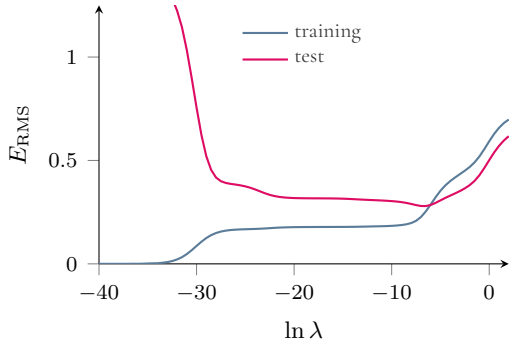


Figure 10: Training and test errors as functions of $\ln \lambda$. The best λ is at the minimum of the test error.

7.4 Bias and Variance

The behavior observed in the example follows from the structure of Equation (16), and is described by the bias–variance decomposition (10), which splits the expected test error into squared bias, variance and the irreducible noise σ^2. When λ is large, the regularization term dominates: the weights are kept small and the predictions nearly flat, whatever the training set, so that different training sets lead to nearly identical models, with low variance, while the systematic patterns of the data are ignored, with high bias. When λ is small, the data term dominates: the model adapts to the noise of each training set, with high variance, while on average it follows the signal, with low bias. Bias thus grows with λ, and variance decreases (Figure 12): the three values of $\ln \lambda$ of the example lie, respectively, where the squared bias dominates, near the minimum of the total error, and where the variance dominates. No choice of λ removes the noise term, a property of the data-generating process, not of the learning algorithm: the best λ minimizes the sum of the other two.

8 Probabilistic Predictors

8.1 From Point Predictions to Distributions

The predictors considered so far are functional: $h(\mathbf{x})$ is a single value, and a loss compares it with the observed target. Two predictions with the same value cannot be told apart in confidence, although the target may be known almost exactly at one input and be very uncertain at another. The foundations notes have introduced the alternative: a probabilistic predictor $h : \mathcal{X} \to \mathcal{D}(\mathcal{Y})$ returns, for each input, a distribution $p_P(\cdot|\mathbf{x})$ over the possible targets. We write $q = h(\mathbf{x})$ for the predicted distribution; in regression q is a density on $\mathbb{R}$. The prediction states not only which

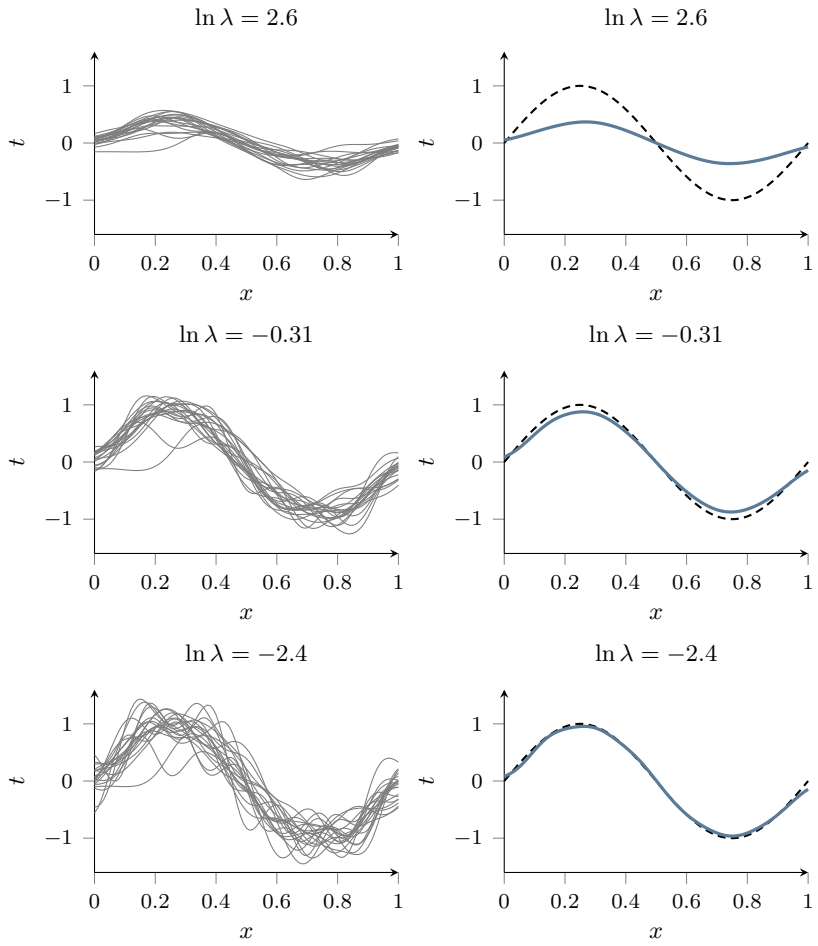


Figure 11: Ridge regression on 100 training sets, for $\ln \lambda = 2.6$ (top), -0.31 (middle) and -2.4 (bottom). Left: twenty of the fitted curves; right: the average of all the 100 curves (blue) and the true function (dashed).

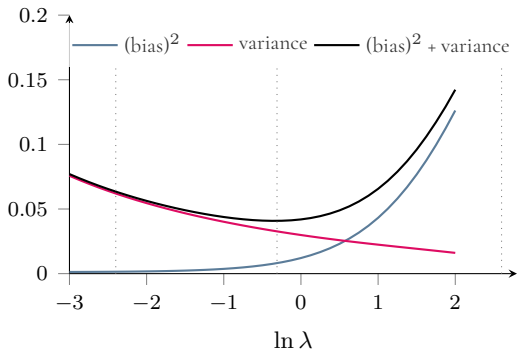


Figure 12: Squared bias, variance and their sum as functions of $\ln \lambda$, estimated from the 100 training sets of Figure 11, whose three values of $\ln \lambda$ are marked by dotted lines. The best λ is at the minimum of the black curve.

values are plausible, but how plausible each of them is: from q one can derive a point prediction, such as its mean or median, but also intervals that contain the target with a given probability, or the probability that the target exceeds a threshold, which is what a decision with asymmetric costs requires.

The definitions of risk, empirical risk and empirical risk minimization of the foundations notes

apply unchanged, with a *scoring rule* in place of the loss. What is specific to the probabilistic case is which scoring rules lead to the right distribution, and why minimizing them on observed values, which are single outcomes and not distributions, achieves this. We discuss these questions in general, and then in the case of regression.

8.2 Comparing Distributions

To evaluate a probabilistic predictor, its prediction q must be compared with the distribution p of the target, the conditional distribution $p_C(\cdot|\mathbf{x})$ at the input considered. Before turning to observed values, it is useful to see two natural ways of comparing two distributions on the real line, which lead to the two scoring rules used in practice (Figure 13, top).

The first is the **cross-entropy** of q relative to p,

$$H(p,q) \;=\; -\int p(u)\log q(u)\,du \;=\; \mathop{\mathbb{E}}_{u\sim p}\left[-\log q(u)\right], \tag{21}$$

the average, over values drawn from the target distribution, of $-\log q(u)$, the *surprise* of the prediction at u: large where the prediction gives a small density to values that occur often. It splits as

$$H(p,q) \;=\; \underbrace{-\int p(u)\log p(u)\,du}_{H(p)} \;+\; \underbrace{\int p(u)\log\frac{p(u)}{q(u)}\,du}_{\mathrm{KL}(p\,\|\,q)}, \tag{22}$$

the *differential entropy* of p, which does not depend on the prediction, plus the **Kullback–Leibler divergence** of q from p. The divergence is non-negative, and zero only for $q = p$. The cross-entropy is therefore minimized, over q, by $q = p$, where it equals the entropy of the target. Both terms require densities: the cross-entropy compares the two distributions value by value, through the ratio of their densities.

The second compares the distribution functions $F_q(u) = \int_{-\infty}^{u} q$ and $F_p(u) = \int_{-\infty}^{u} p$, by the squared L^2 distance between them, the **Cramér distance**

$$D_C(q,p) \;=\; \int \big(F_q(u) - F_p(u)\big)^2 du, \tag{23}$$

the integral of the squared vertical gap between the two curves. It is non-negative, zero only for $q = p$, and symmetric; it is measured in the units of the target, and, since it uses distribution functions rather than densities, it is defined for any pair of distributions, including discrete ones and distributions concentrated on single points.

8.3 Scoring Rules

Both comparisons are **scoring rules**, functions

$$S : \mathcal{D}(\mathcal{Y}) \times \mathcal{D}(\mathcal{Y}) \longrightarrow \mathbb{R}, \tag{24}$$

where $S(q,p)$ is the cost of reporting q when the target is distributed as p; as for losses, lower is better. The definition does not require p to be the distribution of a random target: it can be any distribution, and in particular the distribution δ_t concentrated on a single value, introduced in Section 1. This is the case that matters for learning, since the target distribution is never observed: only single values t are.

8.4 From Distributions to Observed Values

Setting $p = \delta_t$ in the two comparisons gives the two scoring rules used to evaluate a prediction on an observed value (Figure 13, bottom).

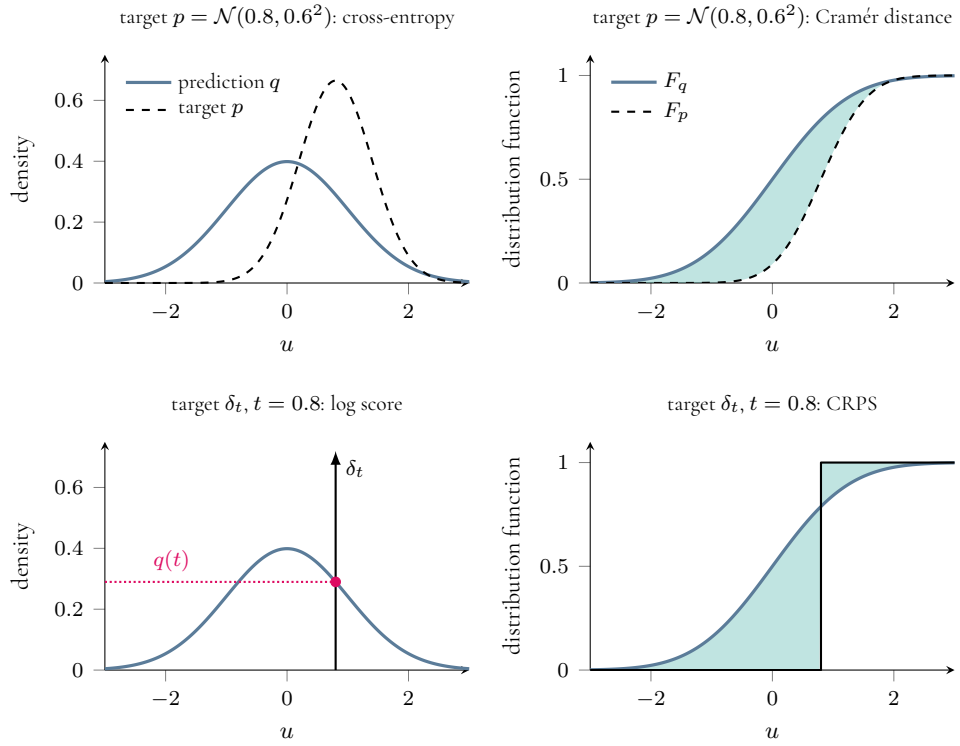


Figure 13: Comparing the prediction $q = \mathcal{N}(0, 1)$ with the target distribution. Top: a target $p = \mathcal{N}(0.8, 0.6^2)$; the cross-entropy compares the densities ($H(p, q) = 1.419$, $H(p) = 0.908$, $\mathrm{KL}(p \,\|\, q) = 0.511$), the Cramér distance is the integral of the squared gap between the distribution functions ($D_{\mathrm{C}} = 0.239$; the absolute gap is shaded). Bottom: the target becomes a single observed value $t = 0.8$, $p = \delta_t$; the cross-entropy reduces to $-\log q(t) = 1.239$, the log score, and the Cramér distance to the CRPS, the integral between F_q and the step at t (0.476).

From the cross-entropy to the log score. The integral in (21) with $p = \delta_t$ is an average over values drawn from δ_t, which are always equal to t, so that

$$S_{\log}(q, \delta_t) \;=\; H(\delta_t, q) \;=\; -\log q(t), \tag{25}$$

the log score: the surprise of the prediction at the value that actually occurred. It charges the predictor according to how much it expected what happened, looking only at the density it gives to the observed value. The decomposition (22) does not survive the limit: δ_t has no density, its differential entropy is $-\infty$, and $\mathrm{KL}(\delta_t \,\|\, q) = +\infty$ for every density q; but their sum, the cross-entropy, remains finite, and is the only part that depends on q.

From the Cramér distance to the CRPS. The distribution function of δ_t is the step $\mathbf{1}[u \geq t]$, and (23) becomes the continuous ranked probability score

$$\mathrm{CRPS}(q, \delta_t) \;=\; D_{\mathrm{C}}(q, \delta_t) \;=\; \int \big(F_q(u) - \mathbf{1}[u \geq t]\big)^2 du, \tag{26}$$

8.5 Proper and Strictly Proper Scoring Rules

A sensible scoring rule must make reporting the true distribution the best strategy on average. S is proper if

$$S(p, p) \;\leq\; S(q, p) \qquad \text{for all } p, q \in \mathcal{D}(\mathcal{Y}), \tag{27}$$

and **strictly proper** if equality holds only for $q = p$, that is, if the truth is the unique minimizer of the expected score,

$$p \;=\; \operatorname*{argmin}_{q \in \mathcal{D}(\mathcal{Y})} S(q, p). \tag{28}$$

For someone who knows that the outcomes are generated by p, $S(p, p)$ is what they pay on average by reporting what they know, and $S(q, p)$ what they pay by reporting anything else: a proper rule never rewards misreporting, a strictly proper one always penalizes it. Without strict propriety, minimizing the risk could lead, reproducibly, to a distribution other than the truth, and more data would not help. The decompositions above show that the log score and the CRPS are strictly proper: the expected score exceeds its value at $q = p$ by a divergence, which vanishes only for $q = p$. In general, $S(q, p) - S(p, p)$ is the divergence associated with a proper rule, and $S(p, p)$ its generalized entropy.

8.6 Risk and Bayes Predictor

At a single input $\mathbf{x}$, the **per-element risk** compares the prediction with the distribution of the target there: $S(h(\mathbf{x}), p_C(\cdot|\mathbf{x}))$ under a probabilistic relation, and $S(h(\mathbf{x}), \delta_{f(\mathbf{x})})$ under a functional one, the special case $p_C(\cdot|\mathbf{x}) = \delta_{f(\mathbf{x})}$. The risk is its average over the items,

$$\mathcal{R}(h) \;=\; \mathop{\mathbb{E}}_{\mathbf{x} \sim p_M} \big[S\big(h(\mathbf{x}), p_C(\cdot|\mathbf{x})\big) \big] \tag{29}$$

The Bayes predictor minimizes the per-element risk at each $\mathbf{x}$, over the predictions a probabilistic predictor can return, which in regression are the densities, ordinary functions $q(u) \geq 0$ with $\int q(u)\, du = 1$. When the target has a conditional density $p_C(\cdot|\mathbf{x})$, the truth is itself an admissible prediction, and for a strictly proper rule

$$h^B(\mathbf{x}) \;=\; p_C(\cdot|\mathbf{x}), \tag{30}$$

with the Bayes risk $\mathop{\mathbb{E}}_{\mathbf{x} \sim p_M} [S(p_C(\cdot|\mathbf{x}), p_C(\cdot|\mathbf{x}))]$, the uncertainty of the target as measured by the scoring rule. Under a functional relation, instead, $\delta_{f(\mathbf{x})}$ is a Dirac delta, not a density, and no predictor can return it. Densities more and more concentrated around $f(\mathbf{x})$ obtain better and better scores, approaching $\delta_{f(\mathbf{x})}$ without reaching it: the per-element risk has an infimum but no minimum, and no Bayes predictor exists.

8.7 Empirical Risk

The conditional distribution $p_C(\cdot|\mathbf{x})$ is never observed, and the training set only provides pairs $(\mathbf{x}_i, t_i)$. Replacing the distribution of the pairs with the empirical distribution of the training set, as for functional predictors, gives the empirical risk

$$\overline{\mathcal{R}}_{\mathcal{T}}(h) \;=\; \frac{1}{|\mathcal{T}|} \sum_{(\mathbf{x},t) \in \mathcal{T}} S\big(h(\mathbf{x}), \delta_t\big), \tag{31}$$

an unbiased estimate of the risk (29) for a fixed h. Each term is the scoring rule evaluated on a point target, the log score or the CRPS of the previous sections: although it compares two distributions, it is computed from the prediction q and the observed value t only. The scoring rule acts as a loss whose first argument is a distribution, and probabilistic ERM is ordinary ERM. With the log score, the empirical risk is the average cross-entropy between the empirical distributions of the observed targets and the predictions.

9 Maximum Likelihood and Empirical Risk Minimization

9.1 Likelihood

The log score connects probabilistic predictors with a classical principle of statistical estimation, maximum likelihood, which we briefly recall. Let $\mathcal{D} = \{z_1, \dots, z_n\}$ be drawn independently from a distribution $p(\cdot\,; \theta)$ of a parametric family, with θ unknown; p is a density for continuous z, a probability mass function for discrete z. The **likelihood** of the dataset is the probability, or density, of the observed data as a function of θ,

$$p(\mathcal{D}; \theta) \;=\; \prod_{i=1}^{n} p(z_i; \theta), \tag{32}$$

where the product follows from independence, and the **maximum likelihood estimator** (MLE) is the value of θ under which the observed data are most probable,

$$\hat{\theta}_{\mathrm{MLE}} \;=\; \operatorname*{argmax}_{\theta}\, p(\mathcal{D}; \theta). \tag{33}$$

The definition applies to any family for which the likelihood can be written: the parameter of a Bernoulli distribution, the mean and variance of a Gaussian, the weights of a neural network.

Example: a coin

Let $z \in \{0, 1\}$ be the outcome of a coin flip, with $p(z; \theta) = \theta^z (1-\theta)^{1-z}$. For $\mathcal{D} = \{1, 1, 0, 1\}$,

$$p(\mathcal{D}; \theta) \;=\; \theta^3 (1-\theta), \qquad \frac{d}{d\theta} \log p(\mathcal{D}; \theta) \;=\; \frac{3}{\theta} - \frac{1}{1-\theta} \;=\; 0 \;\Longrightarrow\; \hat{\theta}_{\mathrm{MLE}} \;=\; \frac{3}{4},$$

the observed proportion of heads.

As a function of $\mathcal{D}$ for fixed θ, $p(\mathcal{D}; \theta)$ is a distribution over datasets; as a function of θ for fixed $\mathcal{D}$, as used by MLE, it is not a distribution over θ, and in general $\displaystyle\int p(\mathcal{D}; \theta)\, d\theta \neq 1$. MLE is an optimization procedure, not a probabilistic statement about θ: a distribution over θ requires a prior, as in the Bayesian approach of Section 12. For the coin of the example, the area under the likelihood is $\int_0^1 \theta^3 (1-\theta)\, d\theta = \frac{1}{20}$ (Figure 14, left).

Since the logarithm is strictly increasing, the MLE also maximizes the **log-likelihood** $\ell(\theta) = \displaystyle\sum_{i=1}^{n} \log p(z_i; \theta)$, a sum, easier to differentiate than a product and numerically safer, since a product of many factors smaller than 1 quickly underflows. Then

$$\hat{\theta}_{\mathrm{MLE}} \;=\; \operatorname*{argmax}_{\theta}\, \ell(\theta) \;=\; \operatorname*{argmin}_{\theta}\, \frac{1}{n} \sum_{i=1}^{n} \big(-\log p(z_i; \theta)\big)\,: \tag{34}$$

MLE minimizes the average log score of the distribution $p(\cdot\,; \theta)$ on the observed data. It is empirical risk minimization with the log score, in which the predicted distribution is the member of the family indexed by θ, and inherits its strict propriety: as n grows, if the true distribution belongs to the family, the minimizer approaches it.

The right panel of Figure 14 shows this for samples of increasing size from a coin with $\theta_0 = 0.6$. The average log score on the sample fluctuates around its expectation, the cross-entropy $-\theta_0 \log \theta - (1-\theta_0) \log(1-\theta)$, which by strict propriety is minimized exactly at θ_0; as n grows the fluctuations shrink, and the minimizer, the observed proportion of heads, moves toward θ_0.

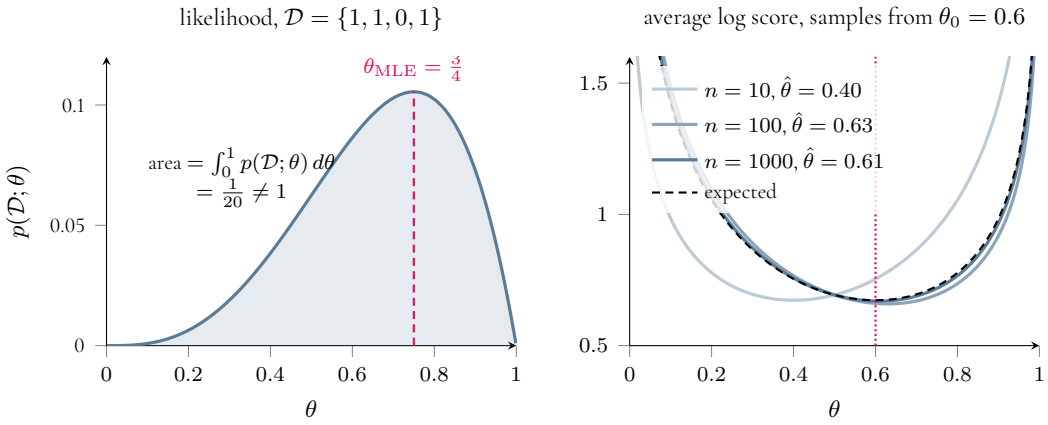


Figure 14: Maximum likelihood for a coin. Left: the likelihood $\theta^3(1-\theta)$ of $\mathcal{D} = \{1, 1, 0, 1\}$, maximized at $\hat{\theta}_{\text{MLE}} = \frac{3}{4}$; its area is $\frac{1}{20}$, so that it is not a density over θ. Right: average log score $-\ell(\theta)/n$ on samples of size 10, 100, 1000 drawn with $\theta_0 = 0.6$ (dotted), and its expectation (dashed); the minimizers, 0.40, 0.63, 0.61, approach θ_0.

9.2 Conditional Models and the Induced Loss

In supervised learning the data are drawn from $p_J(\mathbf{x}, t) = p_M(\mathbf{x})\, p_C(t|\mathbf{x})$. Generative approaches, discussed later in the course, model the whole joint distribution; the **discriminative** approach followed here models only $p_C(t|\mathbf{x})$, which is what prediction requires. A probabilistic predictor with values in a parametric family $\mathcal{P}(\cdot\,; \theta)$ provides, at each input, the parameter of the distribution of the target,

$$t \sim p_C(\cdot|\mathbf{x}) = \mathcal{P}\big(\cdot\,; h(\mathbf{x})\big), \qquad h \in \mathcal{H}, \tag{35}$$

such as its mean, or its mean and variance, in the Gaussian case. Treating the inputs as given, the **conditional likelihood** of the targets is $\prod_{i=1}^{n} \mathcal{P}(t_i; h(\mathbf{x}_i))$, in which p_M does not appear and never needs to be modeled, and maximizing it is the same as minimizing the empirical risk (31) with the log score. When the predictor provides only a location parameter $y = h(\mathbf{x})$, all other parameters of the family, such as the noise level, being fixed, the log score becomes a function of y and t only, the **MLE-induced loss**

$$L\big(h(\mathbf{x}), t\big) = -\log \mathcal{P}\big(t; h(\mathbf{x})\big), \tag{36}$$

and maximum likelihood estimation of h is ERM with a functional loss. Choosing a probabilistic model for the conditional distribution of the target is thus the same as choosing a loss: *which family generated the targets?* and *what should the empirical risk penalize?* have the same answer.

The converse question is whether a given loss is induced by some model, that is, whether there is a family of densities with $-\log \mathcal{P}(t; y) = a\, L(y, t) + c$ for some constants $a > 0$ and c. Such a family, if it exists, is $\mathcal{P}(t; y) \propto e^{-aL(y,t)}$, which requires two conditions: $e^{-aL(y,t)}$ must be integrable in t, and its integral must not depend on y, since otherwise the normalization would add to $-\log \mathcal{P}$ a term depending on the prediction, and the induced loss would differ from L. For a loss depending only on the residual, $L(y, t) = \rho(t - y)$, the second condition always holds, since translating t does not change the integral, and the loss is induced by the **additive noise** model $t = y + \varepsilon$, with noise density $p_N(\varepsilon) \propto e^{-a\rho(\varepsilon)}$, as soon as this is integrable. Gaussian noise induces the quadratic loss, as shown in the next section, Laplace noise the absolute loss, and a hybrid density the Huber loss (Section 15); an asymmetric Laplace density, with different slopes on the two sides, induces the asymmetric absolute loss used to estimate quantiles.

Figure 15 shows these pairs. Each loss is, up to constants, minus the logarithm of the corresponding noise density, so that the shape of one determines the shape of the other: the heavier the tails of the density, the more slowly the loss grows for large residuals. The Gaussian, with the lightest tails,

induces a loss growing quadratically, which heavily penalizes large residuals and is therefore sensitive to outliers; Laplace noise, with heavier tails, makes large residuals more plausible, and induces a loss growing only linearly. The Huber density is Gaussian at the center and Laplace in the tails, and the asymmetric Laplace density, being more spread on one side, induces a loss with different slopes for positive and negative residuals.

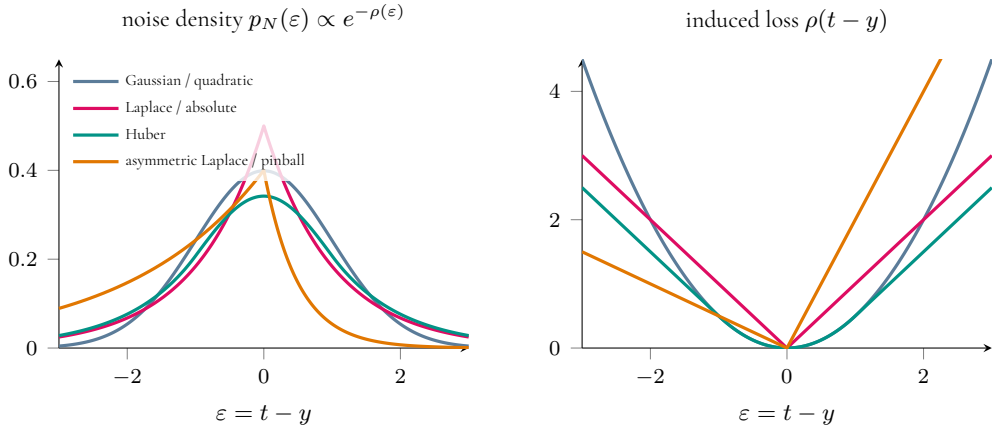


Figure 15: Noise densities (left) and the losses they induce (right), $\rho(\varepsilon) = -\log p_N(\varepsilon)$ up to an additive constant: Gaussian noise and quadratic loss, Laplace noise and absolute loss, Huber (here with threshold 1), and asymmetric Laplace noise with the pinball loss, here with slopes 2 and 0.5 on the two sides.

Other losses are not induced by any model. A *bounded* loss, such as the truncated quadratic loss $\min\big((y-t)^2, c\big)$, used in robust statistics to ignore gross outliers completely, gives e^{-aL} tending to the constant e^{-ac} for large residuals, which is not integrable: no density corresponds to it. A loss whose scale depends on the prediction, such as the relative squared error $(y-t)^2/y^2$, used when errors matter in proportion to the size of the target, leads to a normalization depending on y: the Gaussian with standard deviation proportional to y, the natural candidate, has log score $(y-t)^2/(2s^2y^2) + \log|y| + c$, which contains the additional term $\log|y|$, and is not a multiple of the relative error. In classification, the perceptron loss, as shown in the notes on linear classification, and the hinge loss are not negative log-likelihoods of any model of the labels either: for the hinge loss, for instance, $e^{-\max(0,1-y)} + e^{-\max(0,1+y)}$, the sum over the two labels, is not constant in y.

The next section develops the Gaussian case, estimating by maximum likelihood also the noise level.

10 A Probabilistic Model

The models developed so far are functional: they return a single value for each input, with no indication of how reliable it is, so that two predictions with the same value cannot be told apart in confidence. A probabilistic model treats the target t at a given input as a random variable, whose mean is predicted by the model, while noise, measurement errors and unmodeled factors push each observation away from the mean. This is not a claim that the world is random, but an acknowledgment that the model does not capture every factor affecting the target. Following Section 9, the conditional distribution $p_C(t|\mathbf{x})$ is assumed to belong to a parametric family, whose parameter is provided by the predictor, and the choice of the family determines the loss.

The standard choice is a Gaussian centered at the prediction, with **precision** $\beta = \frac{1}{\sigma^2}$ (Figure 16):

$$p(t|\mathbf{x}; \overline{\mathbf{w}}, \beta) \;=\; \mathcal{N}\big(t|h(\mathbf{x}; \overline{\mathbf{w}}), \beta^{-1}\big). \tag{37}$$

A large β means low noise, with observations tightly clustered around the prediction; a small β means large noise. The Gaussian choice has two justifications. A deviation resulting from many

small independent perturbations is approximately Gaussian, by the central limit theorem; and, among all distributions with a given mean and variance, the Gaussian has the largest entropy, so that it is the least committal choice when there is no reason to prefer a skewed or heavy-tailed noise. The MLE-induced loss of this model is

$$-\log p(t|\mathbf{x};\overline{\mathbf{w}},\beta) \;=\; \frac{\beta}{2}\big(t-h(\mathbf{x};\overline{\mathbf{w}})\big)^2 - \frac{1}{2}\log\frac{\beta}{2\pi}, \tag{38}$$

and, since neither the additive term nor the factor $\beta/2$ depends on h, the quadratic loss of Section 4 is the loss induced by independent Gaussian targets centered at $h(\mathbf{x})$, with constant variance.

Equivalently, $t = h(\mathbf{x}) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0,\beta^{-1})$ independent of $\mathbf{x}$. Replacing the Gaussian with any fixed density p_N gives the **additive noise** model, in which $p_C(t|\mathbf{x}) = p_N(t-h(\mathbf{x}))$ is the noise distribution translated by $h(\mathbf{x})$, and the induced loss is $-\log p_N$ of the residual; the losses of Section 15 correspond to other choices of p_N. The Gaussian model makes three assumptions, which can be checked on the residuals of a fitted model. **Symmetry**: over- and under-prediction are penalized equally, which is inappropriate when they have different costs. **Light tails**: the fast growth of the loss is justified only if large errors are rare. **Constant variance**: all residuals are equally informative, which fails under **heteroscedastic** noise, whose variance $\sigma^2(\mathbf{x})$ changes with $\mathbf{x}$.

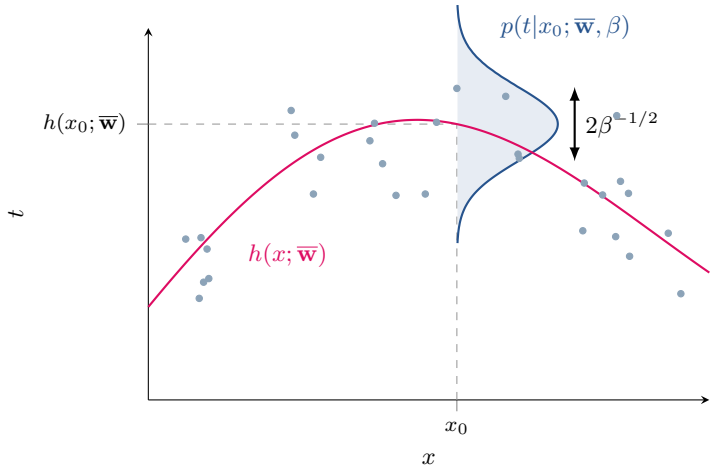


Figure 16: The probabilistic regression model: for each input x, the target t is drawn from a Gaussian centered at $h(x;\overline{\mathbf{w}})$, with variance β^{-1}; the points are a sample generated by the model.

10.1 Maximum Likelihood

The parameters are estimated, as in Section 9, by maximizing the **likelihood** of the training set. For independent items, with residuals $r_i = h(\mathbf{x}_i;\overline{\mathbf{w}}) - t_i$,

$$p(\mathbf{t}|\mathbf{X};\overline{\mathbf{w}},\beta) \;=\; \prod_{i=1}^{n}\mathcal{N}\big(t_i|h(\mathbf{x}_i;\overline{\mathbf{w}}),\beta^{-1}\big) \;=\; \prod_{i=1}^{n}\sqrt{\frac{\beta}{2\pi}}\,e^{-\frac{\beta}{2}r_i^2}, \tag{39}$$

and, taking the logarithm, which turns the product into a sum and has the same maximizer,

$$\log p(\mathbf{t}|\mathbf{X};\overline{\mathbf{w}},\beta) \;=\; -\frac{\beta}{2}\sum_{i=1}^{n} r_i^2 + \frac{n}{2}\log\beta - \frac{n}{2}\log(2\pi). \tag{40}$$

The parameters $\overline{\mathbf{w}}$ appear only in the first term: since $\beta > 0$, maximizing the log-likelihood with respect to $\overline{\mathbf{w}}$ is the same as minimizing the sum of squared residuals. As expected from the induced loss, least squares is the maximum likelihood estimator under Gaussian noise, and $\overline{\mathbf{w}}_{ML}$ is given by

the normal equations (14). Setting to zero the derivative with respect to β,

$$-\frac{1}{2}\sum_{i=1}^{n} r_i^2 + \frac{n}{2\beta} \;=\; 0 \qquad \Longrightarrow \qquad \beta_{ML}^{-1} \;=\; \frac{1}{n}\sum_{i=1}^{n} r_i^2 \,: \tag{41}$$

the estimated noise variance is the mean squared residual of the fitted model.

The result is a **predictive distribution**, a Gaussian probabilistic predictor with constant variance, in the sense of Section 8:

$$p(t|\mathbf{x}; \overline{\mathbf{w}}_{ML}, \beta_{ML}) \;=\; \mathcal{N}\big(t|h(\mathbf{x}; \overline{\mathbf{w}}_{ML}), \beta_{ML}^{-1}\big). \tag{42}$$

Its mean is the point prediction, and $\pm\beta_{ML}^{-1/2}$ gives a one-standard-deviation band around it. The distribution can also be used generatively: sampling t for a collection of inputs produces a synthetic dataset with the same residual distribution and the same spread of the targets as the training data, which is useful to augment a small training set, to test how the results of later processing vary across replicates of the data, and to check whether data generated by the fitted model look plausible.

11 Bayesian Inference

11.1 Prior and Posterior

Maximum likelihood treats the parameters $\boldsymbol{\theta}$ as fixed unknown constants, and the data as the random quantity: this is the *frequentist* view, in which probability is the long-run frequency of outcomes over repetitions of an experiment, and a statement such as "$\theta = 0.5$ with probability 0.3" has no meaning. In the *Bayesian* view, probability is a degree of belief about an uncertain quantity, and any unknown quantity, the parameters included, receives a distribution describing the current state of knowledge about it. What is believed before observing the data is encoded in a **prior distribution** $p(\boldsymbol{\theta})$, which is updated by Bayes' rule, once the training set $\mathcal{T}$ is observed, into the **posterior distribution**

$$p(\boldsymbol{\theta}|\mathcal{T}) \;=\; \frac{p(\mathcal{T}|\boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(\mathcal{T})} \;\propto\; p(\mathcal{T}|\boldsymbol{\theta})\, p(\boldsymbol{\theta}). \tag{43}$$

The posterior combines two sources of information: the likelihood, which favors the parameter values under which the data are probable, and the prior, which disfavors those considered implausible in advance. The denominator $p(\mathcal{T}) = \int p(\mathcal{T}|\boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$, the **evidence**, normalizes the posterior, but does not depend on $\boldsymbol{\theta}$, and plays no role in comparing parameter values. As the number of observations grows, the likelihood dominates and the posterior concentrates around the values supported by the data, whatever the prior.

Figure 17 shows this for the probability ϕ of heads of a coin with true value 0.3, starting from two different priors: a mild one, centered at $\frac{1}{2}$, and one strongly favoring heads, as if heads had been observed nine times out of ten before the experiment. After ten flips the two posteriors still disagree, with modes at about 0.42 and 0.65, since the second prior is worth as many observations as the data. After one hundred flips both are concentrated near the true value, and the prior has become almost irrelevant.

11.2 Maximum a Posteriori

Maximum likelihood fits a single training set, noise included, and with few data it can commit to extreme values: with a single coin flip landing heads, the ML estimate of the probability of heads is 1, a certainty the data cannot support. The **maximum a posteriori** (MAP) estimate replaces the likelihood with the posterior, and selects its mode:

$$\boldsymbol{\theta}_{MAP} \;=\; \operatorname*{argmax}_{\boldsymbol{\theta}}\, p(\boldsymbol{\theta}|\mathcal{T}) \;=\; \operatorname*{argmax}_{\boldsymbol{\theta}}\, \big(\log p(\mathcal{T}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\big). \tag{44}$$

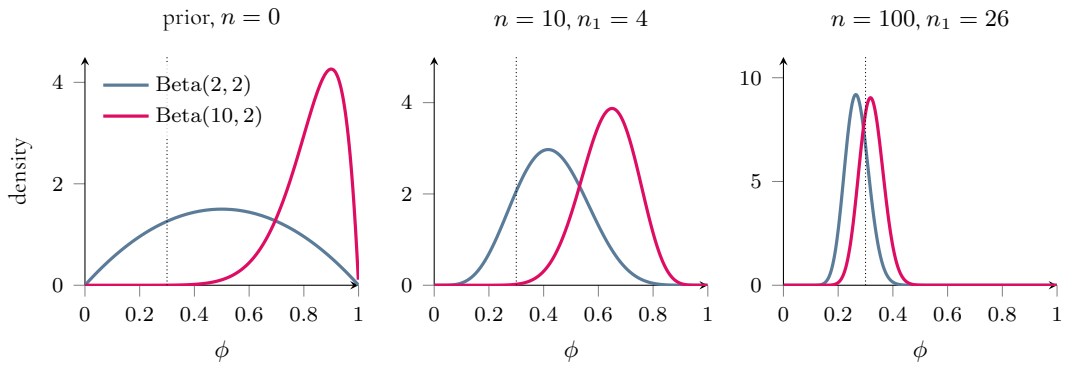


Figure 17: Posterior distributions of the probability ϕ of heads under the priors $\mathrm{Beta}(2,2)$ and $\mathrm{Beta}(10,2)$, for flips of a coin with $\phi_0 = 0.3$ (dotted). Left: the priors. Center: after 10 flips, 4 heads. Right: after 100 flips, 26 heads; the two posteriors nearly coincide.

The logarithm of the prior acts as a penalty added to the log-likelihood. With a zero-mean Gaussian prior $p(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \sigma^2\mathbf{I}) \propto e^{-||\boldsymbol{\theta}||^2/2\sigma^2}$, it is the quadratic penalty $||\boldsymbol{\theta}||^2/2\sigma^2$: ℓ_2 regularization is MAP estimation under a Gaussian prior, with a penalty coefficient equal to the inverse of the prior variance. A narrow prior imposes a strong shrinkage toward zero, a wide one allows larger parameters.

Example: a coin with a Beta prior

For the parameter ϕ of a Bernoulli distribution, a natural prior is the Beta distribution, $p(\phi|\alpha,\beta) \propto \phi^{\alpha-1}(1-\phi)^{\beta-1}$, whose parameters act as *pseudo-counts*: $\alpha - 1$ imaginary successes and $\beta - 1$ imaginary failures, observed before the data; $\alpha = \beta = 2$ expresses a mild belief that the coin is fair. With n_1 successes and n_0 failures in n flips, the posterior is proportional to $\phi^{n_1+\alpha-1}(1-\phi)^{n_0+\beta-1}$, and its mode is

$$\phi_{MAP} = \frac{n_1 + \alpha - 1}{n + \alpha + \beta - 2} :$$

the observed counts are added to the pseudo-counts. With one head in one flip and $\alpha = \beta = 2$, the ML estimate is 1, the MAP estimate $2/3$, which does not commit to certainty on the basis of a single observation.

Figure 18 shows the example on the logarithmic scale. With a single head, the log-likelihood $\log\phi$ keeps increasing up to $\phi = 1$, where ML sets the estimate; the log-prior is a penalty that grows without bound toward both 0 and 1, and adding it moves the maximum to $\frac{2}{3}$. As flips accumulate, the log-likelihood, a sum over the observations, grows in scale while the prior stays the same, and the two estimates become indistinguishable: the prior matters when the data are few, as in the first flips of the sequence shown on the right, which, all tails, give an ML estimate of exactly 0.

11.3 Full Bayesian Inference

MAP still compresses the posterior into a single point, its mode, which summarizes it well when it is symmetric and unimodal, but is misleading when it is skewed or multimodal, and says nothing about the uncertainty that remains. The fully Bayesian approach keeps the whole posterior. A point estimate, if needed, is its mean, $\mathop{\mathbb{E}}_{\boldsymbol{\theta}\sim p(\cdot|\mathcal{T})}[\boldsymbol{\theta}] = \int \boldsymbol{\theta}\, p(\boldsymbol{\theta}|\mathcal{T})\, d\boldsymbol{\theta}$, which weights every value by its posterior probability and reflects the whole shape of the distribution (Figure 19); regions containing a given fraction of the posterior probability, *credible intervals*, quantify the uncertainty. Above all, predictions are made by averaging over the parameters: the predictive distribution of a

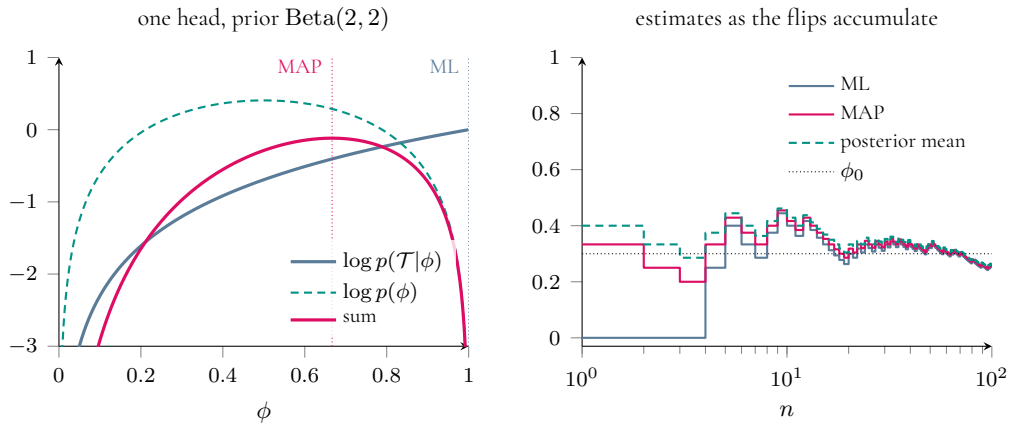


Figure 18: MAP estimation for a coin with prior $\mathrm{Beta}(2,2)$. Left: after one head, the log-likelihood is maximized at $\phi = 1$, the log-posterior, the sum of log-likelihood and log-prior, at $\phi = \frac{2}{3}$. Right: ML, MAP and posterior mean along a sequence of 100 flips with $\phi_0 = 0.3$; with few flips ML takes extreme values, while the estimates using the prior stay moderate, and all three approach ϕ_0 as n grows.

new observation z,

$$p(z|\mathcal{T}) \;=\; \mathop{\mathbb{E}}_{\boldsymbol{\theta}\sim p(\cdot|\mathcal{T})}[p(z|\boldsymbol{\theta})] \;=\; \int p(z|\boldsymbol{\theta})\, p(\boldsymbol{\theta}|\mathcal{T})\, d\boldsymbol{\theta}, \tag{45}$$

averages the predictions of all the parameter values, each weighted by its posterior probability, instead of trusting a single one. ML and MAP are the special cases in which the posterior is replaced by a single point.

The computations are simple when the prior is **conjugate** to the likelihood, that is, when the posterior belongs to the same family as the prior. For the coin, the posterior is the Beta distribution $\mathrm{Beta}(\phi|\alpha+n_1, \beta+n_0)$: each success increments α, each failure β. Its mean, which is also the predictive probability of a success at the next flip, is

$$\mathop{\mathbb{E}}_{\phi\sim p(\cdot|\mathcal{T})}[\phi] \;=\; \frac{\alpha+n_1}{\alpha+\beta+n}, \tag{46}$$

which tends to the ML estimate n_1/n for large n, and is pulled toward the prior mean $\alpha/(\alpha+\beta)$ when the data are few: with one head and $\alpha=\beta=2$ it is $3/5$. The posterior mean along a sequence of flips is shown, dashed, in the right panel of Figure 18. Bayesian linear regression, developed next, is the same construction with a Gaussian prior on the weights, conjugate to the Gaussian likelihood of the linear model.

12 Bayesian Linear Regression

12.1 Prior and Posterior

Maximum likelihood estimates the parameters from the data alone, and with few observations it overfits, since the data do not determine the parameters reliably. The Bayesian approach of Section 11 treats $\overline{\mathbf{w}}$ as a random variable, with a prior distribution updated by Bayes' rule into a posterior distribution. Writing $D = d+1$ for the number of parameters, the standard prior is a zero-mean isotropic Gaussian, with precision α:

$$p(\overline{\mathbf{w}}|\alpha) \;=\; \mathcal{N}(\overline{\mathbf{w}}|\mathbf{0}, \alpha^{-1}\mathbf{I}) \;=\; \left(\frac{\alpha}{2\pi}\right)^{D/2} e^{-\frac{\alpha}{2}\overline{\mathbf{w}}^T\overline{\mathbf{w}}}. \tag{47}$$

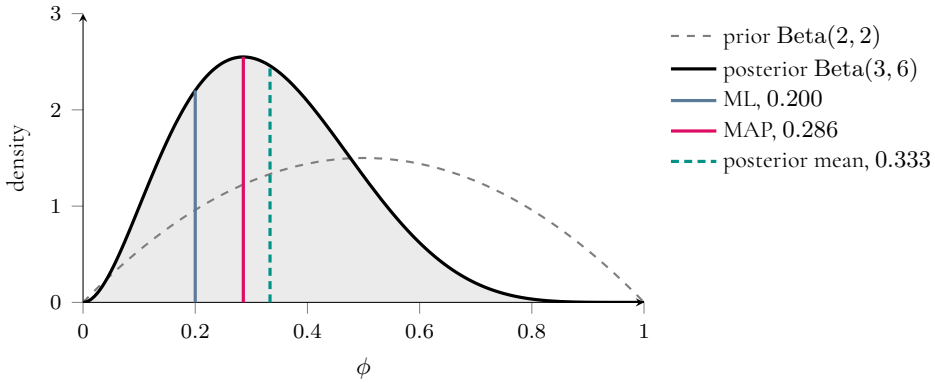


Figure 19: Bayesian inference for the probability ϕ of heads, with prior $\mathrm{Beta}(2, 2)$ and one head in five flips. The posterior $\mathrm{Beta}(3, 6)$ is skewed: the ML estimate (0.2), the MAP estimate, its mode (0.286), and the posterior mean (0.333) are three different summaries of what the data and the prior say about ϕ.

The zero mean expresses no preference for positive or negative weights, and the isotropic covariance treats all directions alike; a larger α concentrates the prior around zero, imposing a stronger preference for small weights (Figure 20).

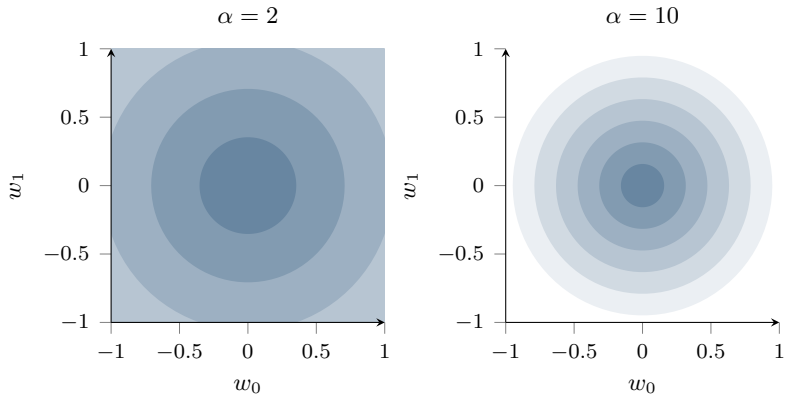


Figure 20: The isotropic Gaussian prior $p(\overline{\mathbf{w}}|\alpha)$ in a two-parameter space, for $\alpha = 2$ and $\alpha = 10$. A large α concentrates it around zero.

By Bayes' rule,

$$p(\overline{\mathbf{w}}|\mathbf{X}, \mathbf{t}; \alpha, \beta) \;\propto\; p(\mathbf{t}|\mathbf{X}, \overline{\mathbf{w}}; \beta)\, p(\overline{\mathbf{w}}|\alpha), \tag{48}$$

where the normalizing constant $p(\mathbf{t}|\mathbf{X}; \alpha, \beta)$ does not depend on $\overline{\mathbf{w}}$. The Gaussian prior is conjugate to the Gaussian likelihood: the posterior belongs to the same family as the prior, which makes all the quantities of interest available in closed form. Expanding the exponents of likelihood and prior, collecting the terms quadratic and linear in $\overline{\mathbf{w}}$ and completing the square, a general Gaussian prior $\mathcal{N}(\overline{\mathbf{w}}|\mathbf{m}_0, \boldsymbol{\Sigma}_0)$ gives a Gaussian posterior $\mathcal{N}(\overline{\mathbf{w}}|\mathbf{m}_p, \boldsymbol{\Sigma}_p)$ with

$$\boldsymbol{\Sigma}_p \;=\; \left(\boldsymbol{\Sigma}_0^{-1} + \beta \overline{\mathbf{X}}^T \overline{\mathbf{X}}\right)^{-1}, \qquad \mathbf{m}_p \;=\; \boldsymbol{\Sigma}_p \left(\boldsymbol{\Sigma}_0^{-1} \mathbf{m}_0 + \beta \overline{\mathbf{X}}^T \mathbf{t}\right). \tag{49}$$

The posterior precision is the sum of the prior precision and of the precision provided by the data, $\beta \overline{\mathbf{X}}^T \overline{\mathbf{X}}$, which grows with every training item; the posterior mean combines the prior mean and the information of the data, weighted by their precisions. For the zero-mean prior,

$$\boldsymbol{\Sigma}_p \;=\; \left(\alpha \mathbf{I} + \beta \overline{\mathbf{X}}^T \overline{\mathbf{X}}\right)^{-1}, \qquad \mathbf{m}_p \;=\; \beta \boldsymbol{\Sigma}_p \overline{\mathbf{X}}^T \mathbf{t} \;=\; \left(\frac{\alpha}{\beta} \mathbf{I} + \overline{\mathbf{X}}^T \overline{\mathbf{X}}\right)^{-1} \overline{\mathbf{X}}^T \mathbf{t}. \tag{50}$$

The posterior mean is the ridge regression solution (19) with $\lambda = \alpha/\beta$: regularization, introduced above as a penalty, is the consequence of a Gaussian prior on the parameters. As $\alpha \to 0$, the prior conveys no information, and the posterior mean tends to the maximum likelihood estimate.

12.2 Sequential Learning

The Bayesian approach processes data incrementally: the posterior obtained from one batch of data is the prior for the next one, so that old data need not be revisited. For batches $\mathcal{T}_1, \mathcal{T}_2, \ldots$,

$$p(\overline{\mathbf{w}}|\mathcal{T}_1, \ldots, \mathcal{T}_k) \;\propto\; p(\mathcal{T}_k|\overline{\mathbf{w}})\, p(\overline{\mathbf{w}}|\mathcal{T}_1, \ldots, \mathcal{T}_{k-1}), \tag{51}$$

and the final posterior is the same as the one obtained from all the data at once, whatever the order of the batches. This is useful when data arrive as a stream, or do not fit in memory.

When a new item $(\mathbf{x}_{n+1}, t_{n+1})$ arrives, the design matrix gains the row $\overline{\mathbf{x}}_{n+1}^T$, so that

$$\overline{\mathbf{X}}_{n+1}^T \overline{\mathbf{X}}_{n+1} \;=\; \overline{\mathbf{X}}_n^T \overline{\mathbf{X}}_n + \overline{\mathbf{x}}_{n+1} \overline{\mathbf{x}}_{n+1}^T$$

and

$$\boldsymbol{\Sigma}_{n+1}^{-1} \;=\; \boldsymbol{\Sigma}_n^{-1} + \beta\, \overline{\mathbf{x}}_{n+1} \overline{\mathbf{x}}_{n+1}^T. \tag{52}$$

The added term is positive semidefinite, since $\mathbf{v}^T \overline{\mathbf{x}}_{n+1} \overline{\mathbf{x}}_{n+1}^T \mathbf{v} = (\overline{\mathbf{x}}_{n+1}^T \mathbf{v})^2 \geq 0$ for every $\mathbf{v}$: each observation adds information, and the covariance can only shrink.

Example: sequential updating

Consider the model $h(x; b, w_1) = w_1 x + b$, with data generated by $t = -0.3 + 0.5\,x$ plus Gaussian noise with variance 0.04 ($\beta = 25$), inputs uniform in $[-1, 1]$, and prior $\mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$ with $\alpha = 2$. Figure 21 shows the posterior and six lines sampled from it, after 0, 1 and 2 observations. Before any observation, the prior is a broad circle centered at the origin, and the sampled lines have arbitrary orientations. After one observation (x_1, t_1), the likelihood is large for all the pairs (b, w_1) whose prediction at x_1 is close to t_1, a stripe in parameter space, and the posterior concentrates along it. A second observation defines a different stripe, and the posterior concentrates near their intersection: the sampled lines all pass close to both points. With more data, the posterior concentrates around the true parameters, whatever the prior.

12.3 Maximum a Posteriori

When a single parameter vector is needed, the **maximum a posteriori** (MAP) estimate is the mode of the posterior,

$$\overline{\mathbf{w}}_{MAP} \;=\; \operatorname*{argmax}_{\overline{\mathbf{w}}} \big(\log p(\mathbf{t}|\mathbf{X}, \overline{\mathbf{w}}; \beta) + \log p(\overline{\mathbf{w}}|\alpha)\big) \;=\; \operatorname*{argmin}_{\overline{\mathbf{w}}} \Big(\frac{\beta}{2} \sum_{i=1}^{n} r_i^2 + \frac{\alpha}{2} \overline{\mathbf{w}}^T \overline{\mathbf{w}}\Big), \tag{53}$$

where the terms not depending on $\overline{\mathbf{w}}$ have been dropped. Dividing by β, this is the ridge objective with $\lambda = \alpha/\beta$, and for a Gaussian posterior the mode coincides with the mean $\mathbf{m}_p$ of Equation (50). The resulting predictive distribution is $\mathcal{N}(t|h(\mathbf{x}; \overline{\mathbf{w}}_{MAP}), \beta^{-1})$. MAP resists overfitting better than maximum likelihood, especially with small training sets, but still relies on a single parameter vector, ignoring the uncertainty that remains about it.

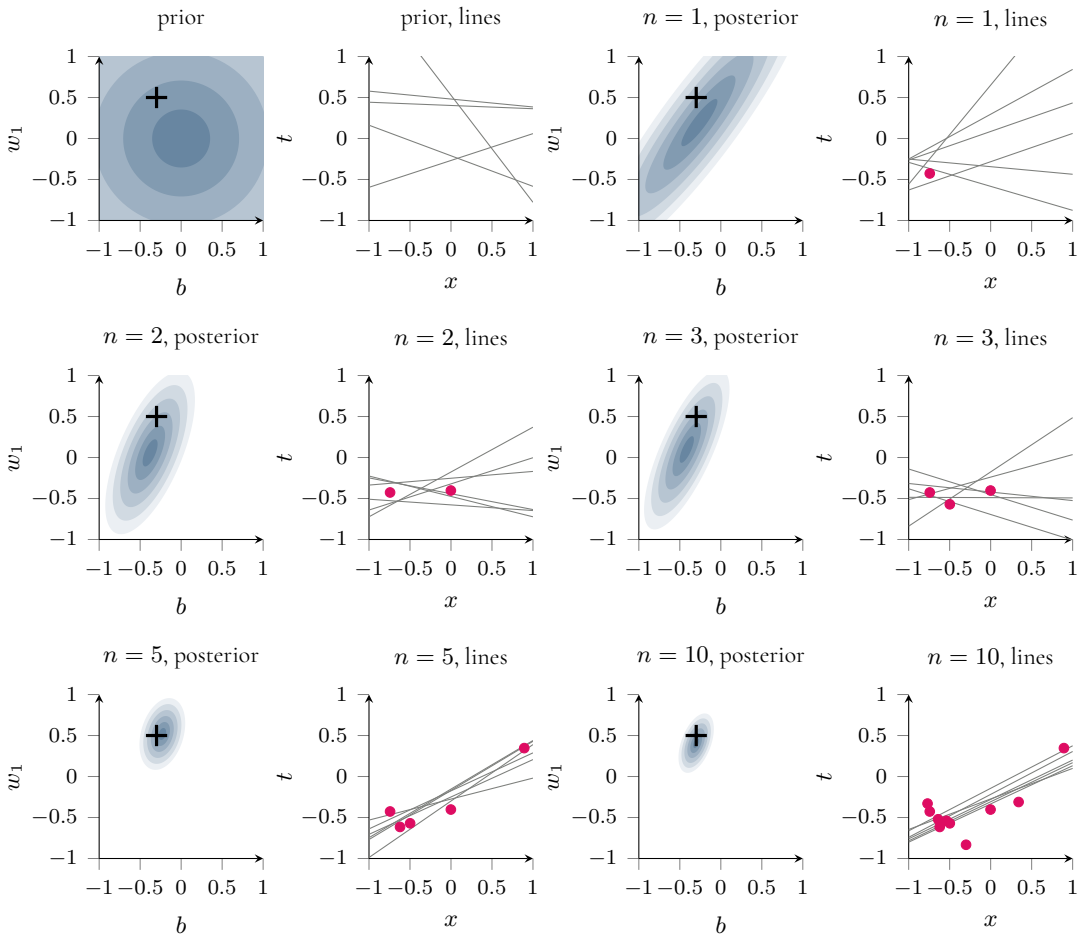


Figure 21: Sequential Bayesian updating for $h(x) = w_1 x + b$. Left: posterior density over (b, w_1) after 0, 1 and 2 observations, with the true parameters marked by a cross; the broad prior collapses into a stripe after one point, and concentrates further after the second. Right: six lines sampled from the current posterior, and the observations (circles); the lines converge toward the true slope 0.5 and intercept -0.3.

13 Fully Bayesian Prediction

13.1 The Predictive Distribution

The fully Bayesian approach uses the whole posterior: the predictive distribution of t at a new input $\mathbf{x}$ averages the predictions of all parameter vectors, weighted by their posterior probability,

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}; \alpha, \beta) \;=\; \int p(t|\mathbf{x}, \overline{\mathbf{w}}; \beta)\, p(\overline{\mathbf{w}}|\mathbf{X}, \mathbf{t}; \alpha, \beta)\, d\overline{\mathbf{w}}, \tag{54}$$

with $p(t|\mathbf{x}, \overline{\mathbf{w}}; \beta) = \mathcal{N}(t|\overline{\mathbf{w}}^T \overline{\mathbf{x}}, \beta^{-1})$ and the Gaussian posterior $\mathcal{N}(\overline{\mathbf{w}}|\mathbf{m}_p, \mathbf{\Sigma})$ of Equation (50), where we write $\mathbf{\Sigma}$ for $\mathbf{\Sigma}_p$. The integral of the product of two Gaussians, one of them depending linearly on the variable of the other, is again Gaussian:

$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}; \alpha, \beta) \;=\; \mathcal{N}\big(t|m(\mathbf{x}), \sigma^2(\mathbf{x})\big), \tag{55}$$

with

$$m(\mathbf{x}) \;=\; \beta \overline{\mathbf{x}}^T \mathbf{\Sigma} \overline{\mathbf{X}}^T \mathbf{t}, \qquad \sigma^2(\mathbf{x}) \;=\; \underbrace{\frac{1}{\beta}}_{\text{noise}} + \underbrace{\overline{\mathbf{x}}^T \mathbf{\Sigma} \overline{\mathbf{x}}}_{\text{parameters}}. \tag{56}$$

The mean is the prediction of the posterior mean, $m(\mathbf{x}) = \mathbf{m}_p^T \overline{\mathbf{x}}$. The variance has two parts. The first, $1/\beta$, is the **aleatoric uncertainty**, the noise of the data, which would remain even if $\overline{\mathbf{w}}$ were known exactly. The second, $\overline{\mathbf{x}}^T \boldsymbol{\Sigma} \overline{\mathbf{x}}$, is the **epistemic uncertainty**, due to the incomplete knowledge of $\overline{\mathbf{w}}$, which shrinks as data arrive. It measures how much of the uncertainty about the parameters projects onto the prediction at $\mathbf{x}$: near the training points, $\overline{\mathbf{x}}$ points in directions in which the data have increased the posterior precision, and the term is small; far from them, it points in directions the data have not constrained, and the term is large. The predictive bands are therefore narrow where the model has seen data and wide elsewhere, the behavior expected of a well calibrated predictor.

Example: evolution of the predictive distribution

Figure 22 shows the predictive distribution for $\sin(2\pi x)$, with 9 Gaussian basis functions, a number that would cause severe overfitting without regularization, after $n = 1, 2, 4$ and 25 observations. With one point, the predictive mean passes through it but is nearly flat elsewhere, the band is wide, and curves sampled from the posterior have very different shapes. With two points, the mean tries to connect them, and the variance remains large far from them. With four points, the shape of the sine wave begins to emerge, and the band narrows at the training points and widens between them. With twenty-five points, the mean closely follows $\sin(2\pi x)$, the band is a thin ribbon, and the sampled curves are nearly indistinguishable: the posterior has concentrated around the best weights.

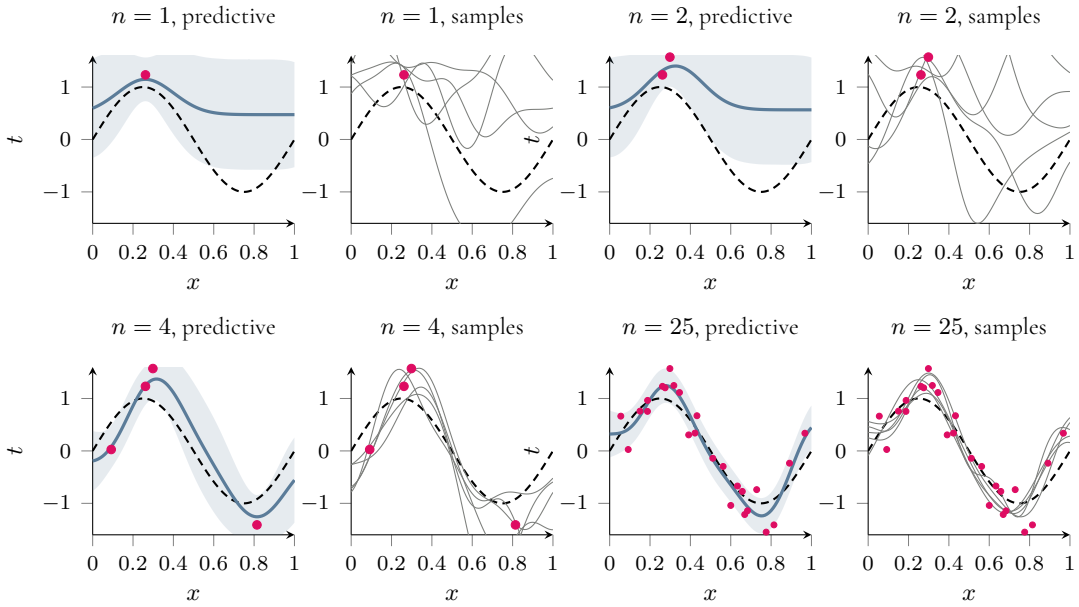


Figure 22: Predictive distribution for $\sin(2\pi x)$ (dashed), with 9 Gaussian basis functions, $\alpha = 2$ and $\beta = 11.1$, after 1, 2, 4 and 25 observations. In each pair of panels, left: training points, predictive mean (blue) and one-standard-deviation band; right: five curves corresponding to parameter vectors sampled from the posterior.

13.2 Three Ways of Predicting

Least squares, MAP and fully Bayesian prediction add, in turn, a layer of probabilistic reasoning (Figure 23 and Table 1). Least squares, equivalent to maximum likelihood, returns a point $\overline{\mathbf{w}}_{LS}$ and a point prediction $\overline{\mathbf{x}}^T \overline{\mathbf{w}}_{LS}$, with no measure of uncertainty. MAP returns a different point, $\overline{\mathbf{w}}_{MAP}$, pulled toward the origin by the prior, equivalent to ridge regression with $\lambda = \alpha/\beta$, and a Gaussian prediction whose variance β^{-1} reflects only the noise, and does not decrease as data

arrive. The fully Bayesian approach uses the whole posterior, and its predictive variance includes the epistemic term, which decreases with the size of the training set and is smaller where the data are denser.

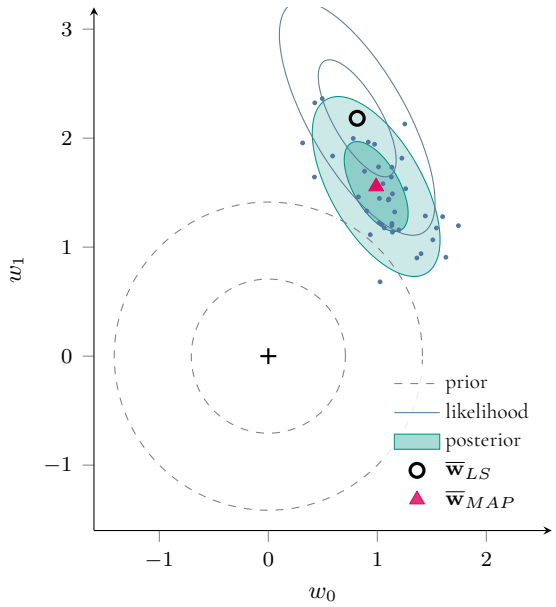


Figure 23: The three approaches in the parameter space of a two-parameter model $h(x) = w_0 + w_1 x$, fitted to four observations. Dashed circles: the prior; solid ellipses: the likelihood, centered at the least-squares estimate $\overline{\mathbf{w}}_{LS}$ (circle); shaded ellipses: the posterior, whose mode is the MAP estimate $\overline{\mathbf{w}}_{MAP}$ (triangle), pulled toward the origin (cross) by the prior. The fully Bayesian approach uses the whole posterior, represented by the dots sampled from it.

	Least squares / ML	**MAP**	**Fully Bayesian**
Learned	$\overline{\mathbf{w}}_{LS}$ minimizing E_D	$\overline{\mathbf{w}}_{MAP}$, posterior mode	posterior $p(\overline{\mathbf{w}}\|\mathbf{X},\mathbf{t})$
Regularization	penalty λ	prior, $\lambda = \alpha/\beta$	prior
Prediction	point $\overline{\mathbf{x}}^T\overline{\mathbf{w}}_{LS}$	$\mathcal{N}(\overline{\mathbf{x}}^T\overline{\mathbf{w}}_{MAP}, \beta^{-1})$	$\mathcal{N}(m(\mathbf{x}), \sigma^2(\mathbf{x}))$
Uncertainty	none	aleatoric	aleatoric and epistemic
Reduces to	—	least squares for $\alpha \to 0$	MAP for $\boldsymbol{\Sigma} \to \mathbf{0}$

Table 1: The three approaches to prediction in linear regression.

14 Choosing the Hyperparameters

14.1 Marginalizing the Hyperparameters

So far, α and β have been fixed in advance. A fully Bayesian treatment would also give them a prior, and marginalize them out:

$$p(t|\mathbf{x},\mathbf{X},\mathbf{t}) = \int p(t|\mathbf{x},\overline{\mathbf{w}};\beta)\, p(\overline{\mathbf{w}}|\mathbf{X},\mathbf{t};\alpha,\beta)\, p(\alpha,\beta|\mathbf{X},\mathbf{t})\, d\overline{\mathbf{w}}\, d\alpha\, d\beta. \tag{57}$$

This integral is not available in closed form, and must be approximated. The Laplace approximation replaces an intractable posterior with a Gaussian centered at its mode $\mathbf{z}_0$, through a second-order expansion of its logarithm, $\ln f(\mathbf{z}) \approx \ln f(\mathbf{z}_0) - \frac{1}{2}(\mathbf{z}-\mathbf{z}_0)^T\mathbf{A}(\mathbf{z}-\mathbf{z}_0)$, with $\mathbf{A}$ the Hessian

of $-\ln f$ at $\mathbf{z}_0$. Variational inference chooses, within a family of simpler distributions, often assuming independent variables, the one closest to the posterior (usually in terms of *Kullback–Leibler divergence*). Markov chain Monte Carlo methods draw samples $(\overline{\mathbf{w}}^{(l)}, \alpha^{(l)}, \beta^{(l)})$ from the posterior, and approximate the predictive distribution by the average $\frac{1}{L}\sum_l p(t|\mathbf{x}, \overline{\mathbf{w}}^{(l)}; \beta^{(l)})$, at a high computational cost. For linear regression, the simpler approach described next is usually sufficient.

14.2 Absolute Loss

The absolute loss is

$$L(y,t) \;=\; |y-t|. \tag{58}$$

A residual r costs $|r|$ instead of r^2: with this linear growth no single large error can dominate the sum, and the loss is much more robust to outliers. Its (sub)gradient is constant, -1 for $y<t$ and $+1$ for $y>t$ (Figure 24), so that outliers and moderate errors receive corrections of the same size. The kink at $y=t$ requires subgradient methods,[2] which converge more slowly than gradient methods on the quadratic loss. The empirical risk is the mean absolute error

$$\overline{\mathcal{R}}_{\mathcal{T}}(h) \;=\; \frac{1}{|\mathcal{T}|}\sum_{(\mathbf{x},t)\in\mathcal{T}} |h(\mathbf{x})-t|, \tag{59}$$

convex for a linear model, so that every local minimum is global, but not strictly convex, so that the minimizer need not be unique.

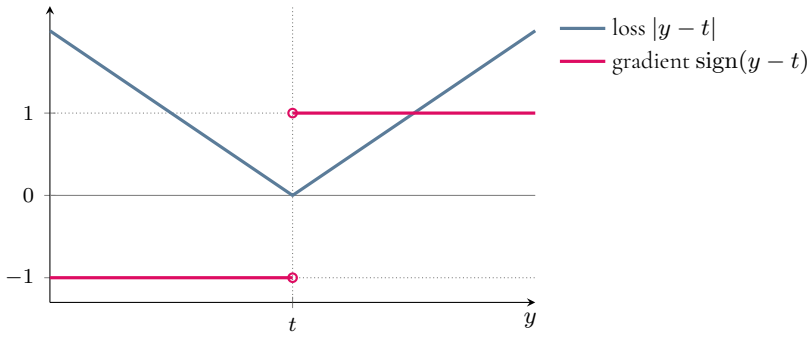


Figure 24: Absolute loss and its (sub)gradient, a constant ± 1 except at the kink $y=t$.

With $P_C(z|\mathbf{x})$ the conditional cumulative distribution, the expected loss of a prediction y and its derivative are

$$g(y) \;=\; \mathop{\mathbb{E}}_{t\sim p_C(\cdot|\mathbf{x})}\left[|y-t|\right], \qquad g'(y) \;=\; P_C(y|\mathbf{x}) - \big(1-P_C(y|\mathbf{x})\big) \;=\; 2P_C(y|\mathbf{x})-1, \tag{60}$$

which vanishes where $P_C(y|\mathbf{x})=1/2$: the Bayes predictor is the conditional median $\nu(\mathbf{x})$. The irreducible risk C is the mean absolute deviation from the median, and plays the role of $\sigma^2(\mathbf{x})$ in the quadratic case. The analogy ends here: the reducible risk $g(h(\mathbf{x}))-g(\nu(\mathbf{x}))$ depends on the whole conditional distribution, and reduces to $|h(\mathbf{x})-\nu(\mathbf{x})|$ only in the FF case, and the bias–variance decomposition loses the simple interpretation of Equation (10).

In the additive noise model, $-\log p_N(\varepsilon)$ is proportional to $|\varepsilon|$, up to a constant, for the Laplace distribution with scale $b>0$,

$$p_N(\varepsilon) \;=\; \frac{1}{2b}\,e^{-|\varepsilon|/b}, \tag{61}$$

[2] At a point where a convex function is not differentiable, any slope between the left and right derivatives is a *subgradient*: for $|r|$ at $r=0$, any value in $[-1,1]$, with 0 chosen by convention.

so that minimizing the mean absolute error is maximum likelihood under independent Laplace residuals. Compared with a Gaussian of the same variance (Figure 25), the Laplace density has a sharp peak at zero and decays exponentially in $|\varepsilon|$ rather than in ε^2: large residuals are uncommon but plausible, and need not be heavily penalized. The same density, used as a prior on the weights rather than as a noise model, makes the MAP estimate the Lasso solution, just as a Gaussian prior gives ridge regression.

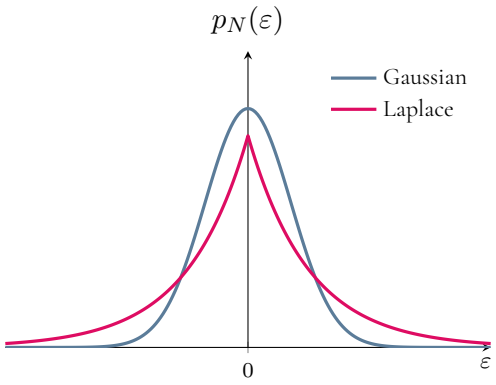


Figure 25: Laplace density with $b = 1$ and Gaussian density with the same variance: the Laplace density has a sharp peak at 0 and heavier tails.

14.3 Huber Loss

The Huber loss combines the smoothness of the quadratic loss with the robustness of the absolute one: it is quadratic for small residuals and linear for large ones, the switch occurring at a threshold $\delta > 0$,

$$L(y,t) \;=\; \begin{cases} \tfrac{1}{2}(y-t)^2 & |y-t| \le \delta, \\ \delta\big(|y-t| - \tfrac{\delta}{2}\big) & |y-t| > \delta. \end{cases} \tag{62}$$

The two pieces join smoothly, so that the loss is convex and differentiable everywhere. Its gradient grows with the residual inside $[-\delta, \delta]$ and is constant at $\pm\delta$ outside (Figure 26): regular items are corrected in proportion to their error, while outliers have a bounded effect. For very large δ the loss coincides with $\frac{1}{2}(y-t)^2$, and for δ close to zero it is approximately $\delta|y-t|$: up to a factor, it ranges from the quadratic to the absolute loss. In practice, δ is set near the typical size of legitimate residuals, estimated from a preliminary fit, or chosen by cross-validation.

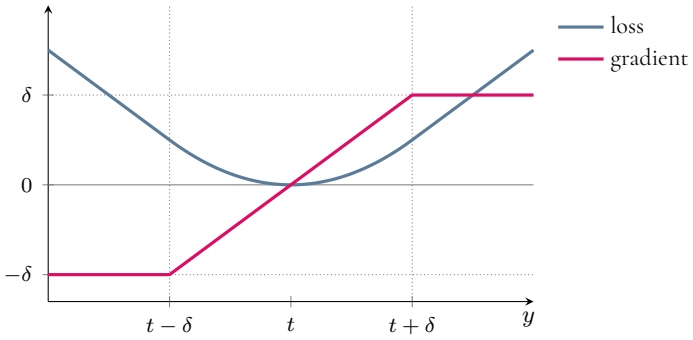


Figure 26: Huber loss and its gradient, linear inside $[-\delta, \delta]$ and constant outside.

The optimal predictor, the Huber estimator, has no closed form. Values of t within δ of the prediction pull it with a force proportional to their distance, as for the quadratic loss, and farther ones with a bounded force, as for the absolute loss: the estimator lies between the median and the mean

(Figure 4), closer to the mean for large δ, and no single observation can have an unbounded influence on it. The corresponding noise model is Gaussian near zero and has Laplace-like tails (Figure 27): small deviations are treated as ordinary measurement noise, large ones as the effect of a different mechanism, such as outliers, data-entry errors or rare events, which a Gaussian model would regard as practically impossible. The threshold δ thus separates ordinary noise from contamination.

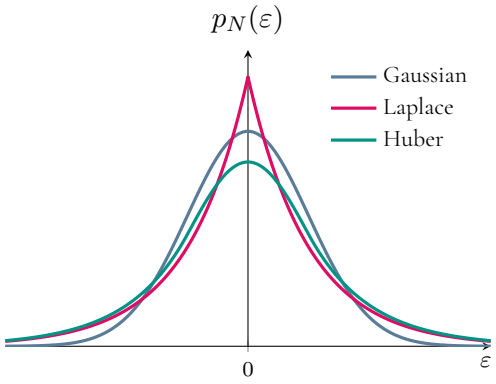


Figure 27: The noise density of the Huber loss for $\delta = 0.5$: Gaussian near 0, with the exponential decay of the Laplace density beyond $|\varepsilon| = \delta$.